THE

SCARA Robot

# SCARA Robot THE600 Ceiling Mount Specifications Manual

(Compatible Controller: TS5000)

Rev. 1

October 1, 2022

Important

- Make sure that this instruction manual is delivered to the final user of SHIBAURA MACHINE's industrial robot.
- Before operating the industrial robot, read through and completely understand this manual.
- After reading through this manual, keep it nearby for future reference.

Shibaura Machine

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# Introduction

Thank you very much for your purchase of SHIBAURA MACHINE's robot system.
This manual describes the basic specifications of the industrial robot, and how to unpack and install the robot, how to connect wiring, how to install the hand, and how to maintain and inspect the robot.
This manual omits content that is identical to the model with standard specifications, and so use this manual together with the instruction manual “STA1171: THE600 Robot Manual”.
Before actually unpacking the machine, be sure to look through the manual once.
Before using this manual, read “STA1179: THE Series Safety Manual” to ensure that you fully understand the implemented safety measures.
This manual has the following chapters.

Specifications
Section 1: Specifications
This section describes the basic specifications and names of respective units of the robot.

Installation
Section 2: Unpacking and Transportation
This section describes how to unpack the robot and how to transport it to the installation site.
This section also describes precautions for temporarily storing the unpacked robot.

Section 3: Installation of the Robot
This section describes the installation environment, required space, and installation procedure for the robot.

Sections 4 and 5: Setting of the Hand
These sections describe how to install the hand on the robot arm and how to wire the hand.
Lastly, these sections describe the permissible load conditions of the hand.

Maintenance
Sections 6: Maintenance
These sections describe how to maintain respective units of the robot.

Section 7: Replacement Parts
This section describes replacement parts of the robot.

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## Use Range

This product is the horizontally articulated type industrial robot in which the manipulator has two horizontally rotating joints and the mechanical interface on its tip moves vertically and rotates horizontally.
The product has been developed with intention of adapting it to automation of wide range of work including conveying work such as movement and alignment of workpieces and assembly work such as insertion of parts and screw tightening.
The customer should also use this product with the aim to adapt it to automation of these work and incorporate it to the automation equipment.

This product is not intended to apply it to work that contacts a human (such as work related to medical care and nursing care), incorporate it to the equipment that supports human life (such as life support device and incubator), incorporate it to the equipment on which a human rides (including transport equipment such as car, aerospace equipment such as airplane, and game equipment such as roller coaster), and incorporate it to the equipment that protects human life and human body (such as safety device).
This product is not intended to incorporate it to the equipment aimed to control large quantity of heat (such as nuclear power controller and incinerator controller).
This product is prohibited from being used to apply it to work or incorporate it to the equipment that is likely to threaten human life and harm a human body directly due to a failure or malfunction. If this product is used, it will be guaranteed under customer's responsibility.

## Use Method

The use conditions for the THE series robot are described. If the product is used in the state that differs from the conditions, unexpected risk may occur. Abide by the conditions for use.

1. Main Specifications of Product: See Specifications "1.4 Specifications Table" and Installation "3.1 Installation Environment."

2. Restrictions on Use: Table 1 and Table 2 give "Restricted Specification Sheet" indicating restrictions on use.

Table 1 Restricted Specification Sheet (1)

| 1. Purpose and application of using product (restrictions on use) | |
|---|---|
| 1) Intended use | Allow the robot to achieve operation along with the purposes (article conveyance, assembly, inspection, machining, etc.).<br>(1) Transportation<br>[1] Secure the transportation route to the unpacking site.<br>[2] Lift the packing box by two or more transportation workers.<br>[3] Transport the packing box to the unpacking site.<br>(2) Unpacking<br>[1] Secure the space required for unpacking.<br>[2] Unpack the box.<br>[3] Make sure that the robot and controller are not damaged.<br>[4] Check the accessories and their quantity.<br>[5] Dispose of or store and manage the packing material.<br>(3) Conveyance<br>[1] Check the posture of the robot during conveyance.<br>[2] Check the mounting of the robot clamp.<br>[3] Check the tightening of the robot clamp set bolts.<br>[4] Remove the bolts that secure the robot base.<br>[5] Lift and convey the robot by two or more workers.<br>(4) Installation<br>[1] Secure the robot to the frame, etc.<br>[2] Remove the robot clamp.<br>[3] Install the controller and connect wiring.<br>[4] Install the hand, etc. |

Table 2 Restricted Specification Sheet (2)

| 1. Purpose and application of using product (restrictions on use) | |
|---|---|
| 1) Intended use (continued) | (5) Teaching<br>- Arrange a person who observes teaching work outside the dangerous range.<br>- Perform work according to the work regulation.<br>[1] Use TP to guide the robot or hold the arm by hand to move the movable part of the robot to any position.<br>[2] Teach the position where the robot operates.<br>[3] Create the operation sequence (program, etc.).<br>[4] Make sure that signal and data are transferred to and from the external equipment.<br>[5] Check operation at low speed (250 mm/sec or less).<br>[6] While increasing the operating speed gradually, make sure that the operation suits the purpose.<br>(6) During operation<br>- A measure has been taken to turn off the servo power when the dangerous range of the robot is approached.<br>- Perform work according to the work regulation.<br>[1] After making sure that there is no person in the fence, enclosure, etc., start operating the robot.<br>[2] Make sure that the operation is performed according to the purpose at the specified speed.<br>[3] Stop operating the robot.<br>(7) Repacking<br>[1] Remove the hand, etc.<br>[2] Mount the robot clamp.<br>[3] Remove the wiring.<br>[4] Pack the robot and controller. |

Table 3 Restricted Specification Sheet (3)

| 1. Purpose and application of using product (restrictions on use) | |
|---|---|
| 1) Intended use (continued) | (8) Maintenance<br>- Arrange a person who observes maintenance work outside the dangerous range.<br>- Perform work according to the work regulation.<br>[Work performed when the primary power is off]<br>[1] Check the bolts for looseness and backlash and re-tighten them.<br>[2] Make sure that the cable clamp is not loose.<br>[3] Check for damage.<br>[4] Clean dust.<br>[5] Inspect and apply antirust agent.<br>[6] Check the tension of the timing belt and presence/absence of crack.<br>[7] Check presence/absence of wear on the cable and air tube.<br>[8] Replace the consumable parts for the robot and controller.<br>[9] Replace the battery for memory in the controller.<br>[Work performed when the primary power is on]<br>[1] Check the grease state and apply grease (when the servo power is off).<br>[2] Check the behavior when the robot is moved by jog guiding using the TP (when the servo power is on).<br>[3] Push each axis by hand and check it for backlash (when the servo power is on).<br>[4] Check abnormal vibration and sound during operation (when the servo power is on).<br>[5] Check the teaching point for misalignment (when the servo power is on).<br>[6] Replace the battery for robot position detector (when the servo power is off).<br>(9) Disposal<br>[1] Mount the clamp on the robot and dispose of the robot using the method defined by each country's laws and regulations. |
| 2) Predictable misuse | - Use for work which applies excessive impact to the robot and to which excessive impact is applied by the robot<br>- Modification of robot, controller, and optional parts and energization with these covers removed<br>- Operation in other than the specified installation state (floor, suspension, and wall-mounted)<br>- Operation on the frame, etc. that do not satisfy the specified strength<br>- Operation under other than the specified installation condition<br>- Entry into the dangerous range of the robot on the assumption that stop for operation signal wait during operation is operation stop<br>- Use under the condition that exceeds the specification value or use in improper setting<br>- Stop of the robot caused by power off during operation and when the servo power is on<br>- Unintended operation caused by erroneous pressing of the button<br>- Operation without checking operation in teaching mode<br>- Operation without checking tightening of the robot and hand set bolts<br>- Unintended operation caused by erroneous wiring<br>- Unintended operation caused by erroneous program selection<br>- Riding on or leaning against the robot or hanging on the arm |

Table 4 Restricted Specification Sheet (4)

| 1. Purpose and application of using product (restrictions on use) | |
|---|---|
| 2) Predictable misuse | - Use for work which applies excessive impact to the robot or to which excessive impact is applied by the robot<br>- Modification of robot, controller, and optional parts and energization with these covers removed<br>- Operation in other than the specified installation state (floor, suspension, and wall-mounted)<br>- Operation on the frame, etc. that do not satisfy the specified strength<br>- Operation under other than the specified installation condition<br>- Entry into the dangerous range of the robot on the assumption that stop for operation signal wait during operation is operation stop<br>- Use under the condition that exceeds the specification value or use in improper setting<br>- Stop of the robot caused by power off during operation or when the servo power is on<br>- Unintended operation caused by erroneous pressing of the button<br>- Operation without checking operation in teaching mode<br>- Operation without checking tightening of the robot and hand set bolts<br>- Unintended operation caused by erroneous wiring<br>- Unintended operation caused by erroneous program selection<br>- Riding on or leaning against the robot or hanging on the arm |
| 3) Unexpected start | - Malfunction caused by excessive noise<br>- Robot start caused by start signal unexpectedly sent from the peripheral equipment to the robot<br>- Unintended start caused by abnormal communication data<br>- Malfunction caused by voltage fluctuation<br>- Power on caused by energization after power outage<br>- Erroneous start caused by failure |

| 2. Replacement, etc. of program components (restrictions on time) | |
|---|---|
| 1) Mechanical restriction | - Before operating the robot, perform operation at a low speed (20% or less of maximum speed) for five (5) minutes<br>- Replace consumable parts as required or periodically through daily inspection, regular inspection (every 3 and 6 months and every year) and overhaul (every 5 years). |
| 2) Electrical restriction | |

Table 5 Restricted Specification Sheet (5)

<table>
<tr><td colspan="3">3 Movable range of product (restriction on space)</td></tr>
<tr><td>1)<br>Operating range</td><td>Axis 1<br>Axis 2<br>Axis 3<br>Axis 4<br>Additional axis<br>-<br>-</td><td>Conforms to 1.4 Specifications Table<br>Conforms to 1.4 Specifications Table<br>Conforms to 1.4 Specifications Table<br>Conforms to 1.4 Specifications Table<br>Conforms to individual specifications.<br>-<br>-</td></tr>
<tr><td>2)<br>Interface</td><td colspan="2">Transportation: The packet robot shall be able to be conveyed by transportation workers.<br>Unpacking: The packet robot shall be able to be carried in.<br>Space of at least 2 $m^3$ shall be secured.<br>Conveyance: The robot shall be able to be lifted up and conveyed by two or more workers.<br>There shall be no obstacle on the route from the unpacking site to the installation site.<br>Installation: The robot shall be able to be moved from the unpacking site to the installation site.<br>The product shall be installed at a location suitable for outer shape and working envelope of the robot.<br>The product shall be installed at a location suitable for outer shape and ventilation of the controller.<br>The cable shall be installed at a location where it is not stepped on by a worker.<br>Teaching: Teaching work shall be able to be performed from outside the dangerous range of the robot.<br>Space shall be secured to prevent a worker from being caught in the robot and peripheral equipment if an entry is made into the dangerous range.<br>Operation: A fence or enclosure shall be installed to prevent any worker from approaching the dangerous range of the robot.<br>The robot shall be able to be operated from outside the dangerous range of the robot.<br>A worker in the fence or enclosure shall be visible from the operation site.<br>Repacking: Work shall be able to be performed to remove the hand, mount the clamp, separate the robot and controller, and remove the robot and controller from the frame.<br>The robot shall be able to be moved from the installation site to the packing site.<br>Maintenance: Space required for maintenance shall be secured.<br>Operation after maintenance shall be able to be checked from outside the dangerous range of the robot.<br>In the dangerous range, space shall be secured to prevent a worker from being caught in the robot and peripheral equipment.<br>Disposal: Work shall be able to be performed to remove the hand, mount the clamp, separate the robot and controller, and remove the robot and controller from the frame.<br>The robot shall be able to be moved from the installation site to the disposal location.</td></tr>
<tr><td>3)<br>Work environment</td><td colspan="2">- The fence, enclosure, etc. surrounding the dangerous range of the robot shall be prepared to prevent any person to make an entry.<br>- A display indicating the operation state of the robot shall be prepared to enable a worker to know that the robot is being operated or stopped.<br>- Illumination shall be prepared to enable a worker to be visually checked.<br>- The floor shall not be covered with oil, etc. to prevent a worker from slipping.<br>- There shall be no noise that impairs attention.</td></tr>
</table>

Table 6 Restricted Specification Sheet (6)

| 4. Life cycle of product |
|---|
| Among the life cycle of assembly, shipment test, transportation, unpacking, conveyance, installation, teaching, operation, repacking, maintenance and disposal, risk assessment is carried out in the life cycle of “conveyance” (transportation, unpacking, conveyance, installation, and repacking) and “use” (teaching, operation, and maintenance), and “disposal.” |

| 5. Person who approaches the machine | | |
|---|---|---|
| Target | Knowledge and condition | |
| 1) Transportation worker | Knowledge | Contents of the instruction manual |
| | Condition | Person who has received training, etc. for work with the conveyance equipment defined by each country’s laws and regulations and to whom work has been permitted by the employer |
| 2) Unpacking worker | Knowledge | Contents of the instruction manual |
| | Condition | - |
| 3) Conveyance worker | Knowledge | Contents of the instruction manual |
| | Condition | - |
| 4) Installation worker | Knowledge | Contents of the instruction manual |
| | Condition | - |
| 5) Teaching worker Work observer | Knowledge | Contents of the instruction manual, operation method for target robot, work regulation, and operation of the robot |
| | Condition | Person who has received training, etc. for robot teaching work defined by each country’s laws and regulations and to whom work has been permitted by the employer |
| 6) Operation worker | Knowledge | Contents of the instruction manual, operation method for target robot, work regulation, and operation of the robot |
| | Condition | - |
| 7) Repacking worker | Knowledge | Contents of the instruction manual |
| | Condition | - |

Table 7 Restricted Specification Sheet (7)

| 5. Person who approaches the machine | | |
|---|---|---|
| Target | Knowledge and condition | |
| 8) Maintenance worker<br>Work observer | Knowledge | Contents of the instruction manual, operation method for target robot, work regulation, and operation of the robot<br>Storage method for target robot |
| | Condition | Person who has received training, etc. for robot inspection work defined by each country's laws and regulations and to whom work has been permitted by the employer |
| 9) Disposal worker | Knowledge | Contents of the instruction manual and waste treatment method |
| | Condition | - |
| 10) Nearby pedestrian<br>Other worker | - Measures shall be taken to prevent entry into the fence, enclosure, etc. surrounding the dangerous range of the robot.<br>- Measures shall be taken to allow the robot to stop operation when the dangerous range is approached.<br>- If other worker enters the dangerous range of the robot for work, robot operation shall be stopped to secure safety under observation by the teaching worker or maintenance worker. | |

| 6. Other |
|---|
| |

# Warranty

We conduct a strict tests and inspections before delivery to ensure that the product satisfies our performance standards.
We provide warranty to cover defects of the product based on the following prerequisites.

1. Warranty period The warranty period of the product is either of the following, whichever is earlier.
   1) 18 months after delivery from our factory
   2) 12 months after installation at your site

2. Warranty scope
   1) We provide warranty for the product. Note that only the specifications and functions defined in the product specifications, catalog, and instruction manual are covered by the warranty. Any secondary or incidental damages caused by failure of the product are not covered by the warranty in any circumstances.
   2) We provide repair of a faulty product for free only if the product has been handled and used according to the instruction manual that comes with the product.
      Any repairing after the warranty period passes will be charged.

3. Disclaimer The warranty shall not apply to the following cases.
   a) Any failure or damage arising from your careless use or misuse that is not defined in the instruction manual.
   b) Any failure caused by aging or normal wear and tear (discoloration, deterioration of consumables*1).
   c) Any failure regarding sensuous phenomena such as noise not affecting the functions
   d) When any modification or disassembly without our consent is confirmed
   e) Any failure or damage arising from insufficient maintenance and inspection or inappropriate repair
   f) Any failure or damage due to act of God, fires, and other force majeure
   g) Internal data such as programs and points created or modified by users
   h) When a product purchased in Japan is taken abroad

4. Precautions
   1) SHIBAURA MACHINE shall not guarantee the standard performance of the product if you use the product beyond the specifications.
   2) If the customer does not observe the “DANGER,” “WARNING,” and “CAUTION” described in this instruction manual, SHIBAURA MACHINE will not assume responsibility for any consequential accident resulting in injury or death, damage or trouble.
   3) “DANGER,” “WARNING,” “CAUTION” and other descriptions stipulated in this instruction manual are only those which can be assumed by SHIBAURA MACHINE without predicting all events in all situations. Please be aware that the range is limited.

*1): Consumable parts refer to the replacement parts for maintenance as listed in instruction Manual “8.2 List of Replacement Parts – Robot.”

## Standards, laws and regulations regarding safety of industrial robots

The international industrial standard ISO/DIS 12100 "Safety of machinery" defines measures for reducing risks regarding machinery.

Step 1: Fundamental safe design measures - Restrictions on force, speed, and energy

Step 2: Safety protection - Installation of protective fences, etc.

Additional protection measures - Installation of emergency stop devices

Step 3: Information for use - Warning labels, alerts, instruction manuals

Based on the above, the ISO/IEC defines the "Guide 51" that groups various standards in a hierarchal structure.

Safety standards for industrial robots are standardized as separate safety standards for machinery in layer C.

- ISO 10218, ISO 10218-1

After risk assessment is carried out, provision of residual risk information to the user is standardized.

- IEC 82079-1

Comply with the standards, laws and regulations in the countries where you use industrial robots.

## Safety Precautions

This manual contains important information for the safe use of the robot body and the prevention of injury to the operator and others and the prevention of damage to property.
Read and understand the meanings of the following indications and symbols first, and observe these precautions.

### [Meanings of Indications]

| Indication | Meaning |
|---|---|
| ⚠ DANGER | Indicates the imminent danger that incorrect operation could cause death or serious injury. |
| ⚠ WARNING | Indicates the danger that incorrect operation could cause death or serious injury. |
| ⚠ CAUTION | Indicates the possibility that incorrect operation could cause injury[1)] to operator. |
| NOTICE | Indicates the possibility that incorrect operation could cause property damage[2)] and advice about performing work. |

1) Injury indicates injuries such as wounds, burns, and electric shock that do not require hospitalization or long-term outpatient treatment.
2) Damage to property indicates extensive damage related to the destruction of assets or materials.

### [Meanings of symbols]

| Symbol | Meaning |
|---|---|
|  | Indicates a prohibited action.<br>The specific content of the prohibition is shown pictorially or in text form nearby the symbol. |
|  | Indicates an instruction (a required action).<br>The specific content of the instruction is shown pictorially or in text form nearby the symbol. |
|  | Indicates a caution and warning (action to which attention is paid).<br>The specific content of the caution is shown pictorially or in text form nearby the symbol. |

**NOTICE**

- In order to install and dispose of the robot safety, be sure to read the “Safety Manual” Instruction Manual provided separately thoroughly before actually starting work.

## Service Network

Please refer to the materials distributed from the SHIBAURA MACHINE agent.

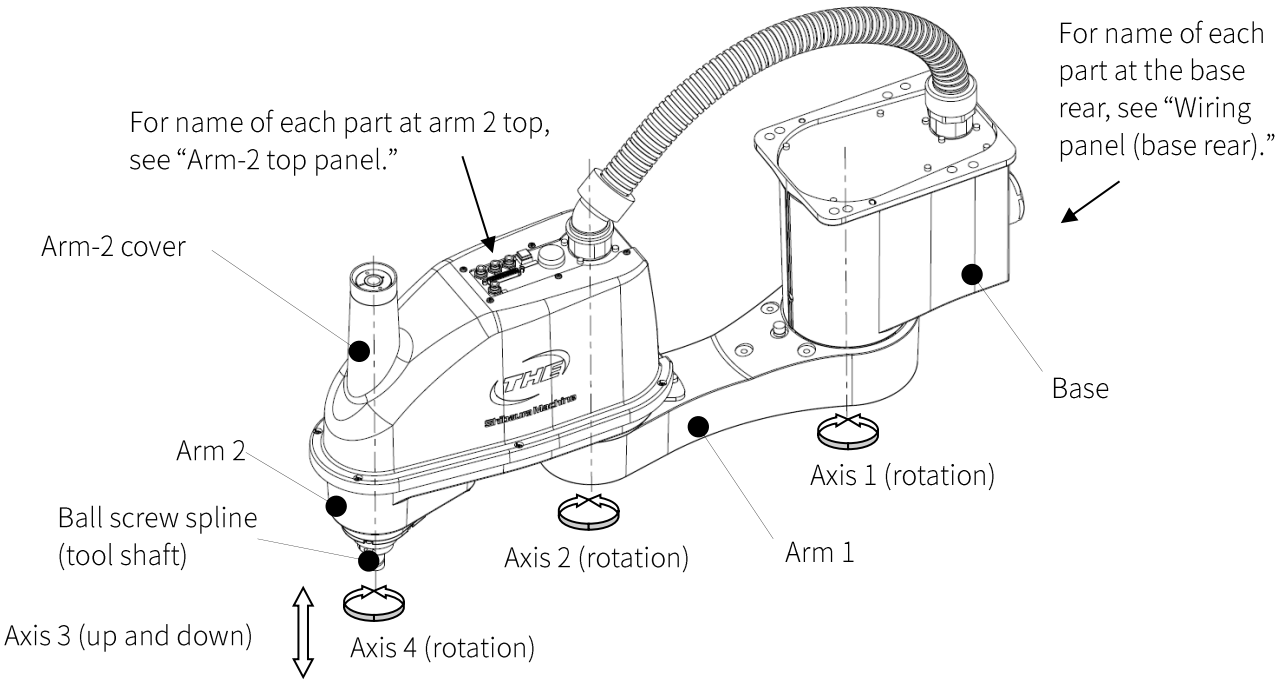
# Specifications

This part describes the basic specifications, name of each part, coordinate system, and external dimensions of the robot.

# 1. Specifications

## 1.1 Name of Each Part

Fig. 1.1 shows the name of each part of the robot.



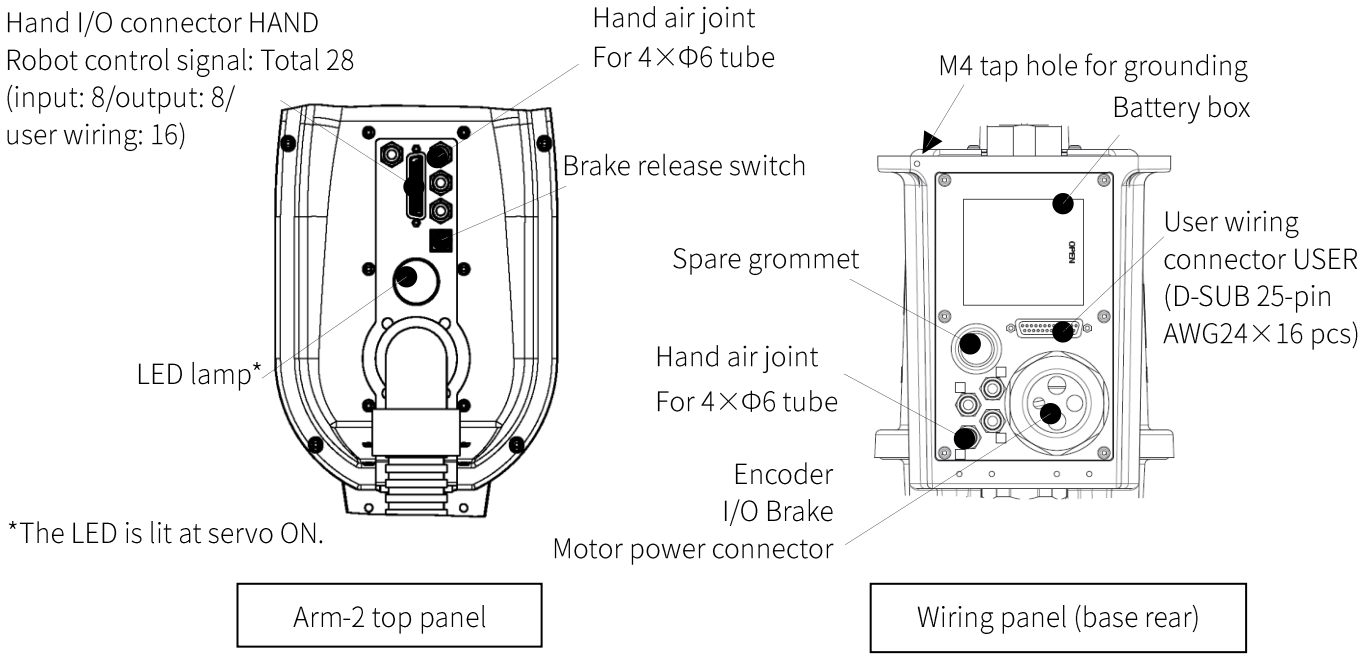




Fig. 1.1 Name of each part

## 1.2 Coordinate System

This is identical to the standard model. For details, see “1.2 Coordinate System” in the “STA1171: THE600 Robot Manual”.

## 1.3 External Dimensions

Fig. 1.2 refers to the external dimensions of the THE600-T.

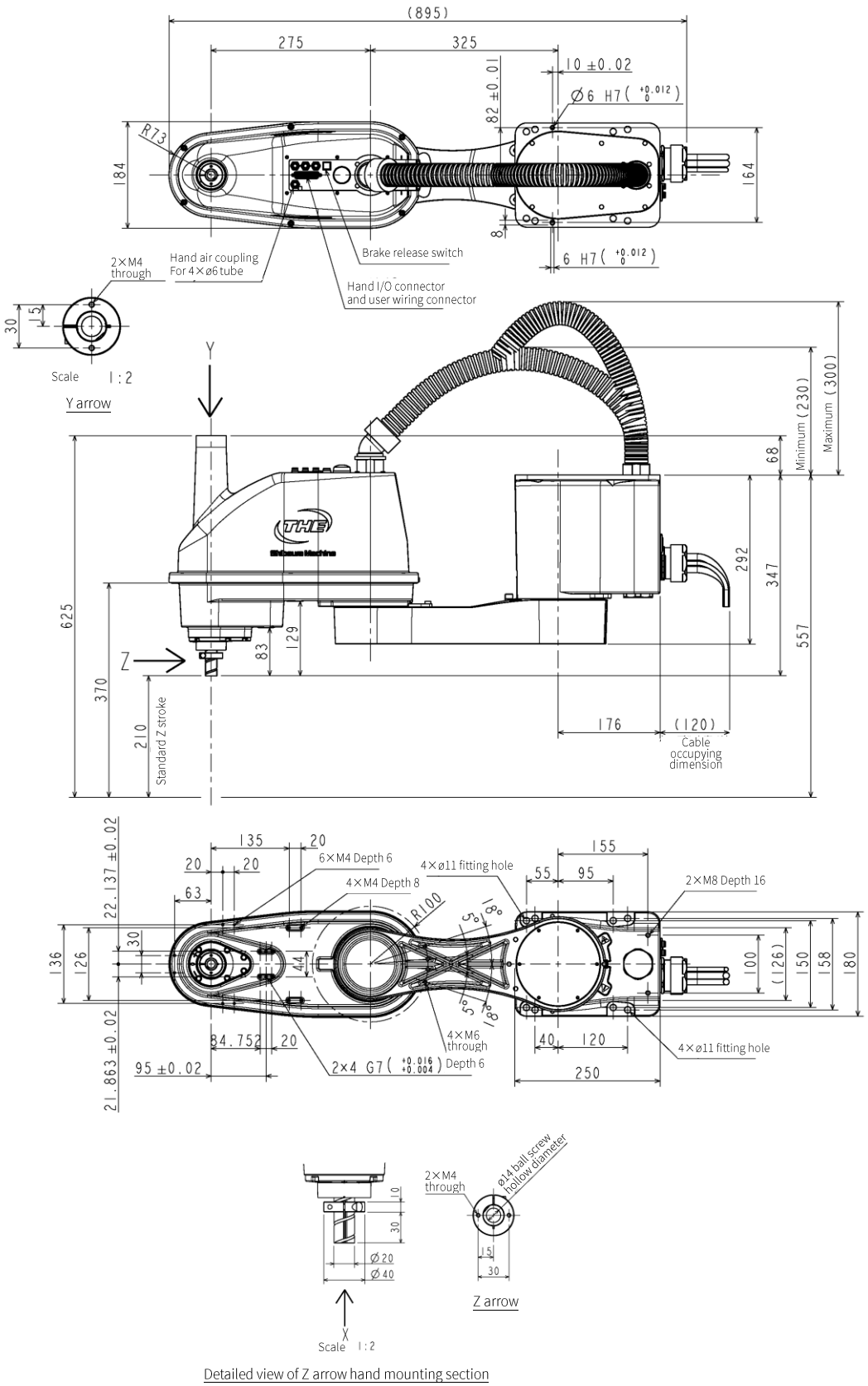



Fig. 1.2 External dimensions of the robot (THE600)

## 1.4 Specifications Table

| Item | | Specifications |
|---|---|---|
| Structure | | Horizontally articulated type SCARA robot |
| Model | | THE600-T |
| Applicable controller | | TS5000 |
| Mass of robot body | | 31 kg |
| Number of controlled axes | | Four (4) |
| Arm length | | 600 mm (275 mm + 325 mm) |
| Motor capacity/current limit | Axis 1 | 1000 W / 28.2Ao-p |
| | Axis 2 | 400 W / 14.1Ao-p |
| | Axis 3 | 400 W / 18.4.Ao-p |
| | Axis 4 | 400 W / 14.1Ao-p |
| Operating range | Axis 1 | ±140 deg |
| | Axis 2 | ±150 deg |
| | Axis 3 | 0 to 210 mm |
| | Axis 4 | ±360 deg |
| Maximum speed *1 | Axis 1 | 457 deg/s |
| | Axis 2 | 672 deg/s |
| | Axis 3 | 2000 mm/s |
| | Axis 4 | 2359 deg/s |
| | Composite speed of axes 1 and 2 | 8.0 m/s |
| Rated payload mass*1 | | 2 kg |
| Maximum payload mass*1 | | 12 kg |
| Permissible load inertia*1 | | 0.25 kg·m$^2$ |
| Maximum offset of load gravity center | | 100 mm |
| Repeatability*2 | Composite speed of axes 1 and 2 | ±0.01 mm |
| | Axis 3 | ±0.01 mm |
| | Axis 4 | ±0.005 deg |
| Cycle time (when payload mass is 2 kg)*3 | | 0.31 sec |
| Drive system | | AC servo motor |
| Position detection method | | Absolute |
| Wiring for hand | | Input: 8, output: 8 |
| Paint color*4<br>Material color*4 | | Arm 2: PANTONE 293C or equivalent<br>Base and arm 1: White AN-90 or equivalent<br>Arm cover: White AN-90 or equivalent |
| Noise level*5 | | 73 dB (A) |
| Power supply capacity*6 | | 4.3 kVA |

*1: The speed and acceleration are limited in accordance with the operation pattern, the load mass, and the offset value.

*2: Repeatability is the repeat accuracy in one direction at a constant ambient temperature and machine temperature. It is not absolute positioning accuracy. It may exceed the specification values depending on the operation pattern, load weight, and offset value. As for X-Y and C, they represent values at the upper limit of Z. Trajectory accuracy is not guaranteed.

*3: Continuous operation with standard cycle operation pattern and exceeding the effective load ratio cannot be performed.
During shuttle for rough positioning in horizontal direction of 300 mm and vertical direction of 25 mm.

*4: The color may differ according to each production lot. Be aware that there is no problem in the very nature of the product quality.

*5: The measurement conditions are as follows.

| | |
|---|---|
| Operating condition: | Rated load, concurrent operation of axes 1 and 2, maximum speed, and maximum acceleration |
| Measurement position: | Height of 1.6 m from the robot installation surface at a distance of 1 m from four front, rear, right and left directions of the robot |

*6: The measurement conditions are as follows.

| | |
|---|---|
| Operating condition: | Rated load and 2 kg, concurrent operation of all axes, maximum speed, and maximum acceleration |
| Measuring instrument: | Power analyzer (POWER ANALYZER 3390) |

| NOTICE | |
|---|---|
| **Prohibited** | • Never use the robot in a high-temperature environment.<br>Otherwise, the performance and lifespan of the alkaline battery used for motor position detector battery are reduced.<br>- If using the robot in a high-temperature environment, please consult with SHIBAURA MACHINE sales office.<br>• Never allow the lowered ball screw spline shaft to collide with obstacles.<br>Otherwise, the ball screw spline shaft will be damaged before an alarm is generated. |
| **Required** | • When micro vibration occurs, be sure to decrease the speed and decelerate the robot.<br>Otherwise, the performance and lifespan of the robot may be reduced. The robot may become faulty.<br>- Use the ACCEL/DECEL command to adjust the speed to the acceleration/deceleration at which the robot does not vibrate.<br>- Micro vibration may occur on the robot tip depending on the posture of the robot.<br>• To operate axes 1, 2, and 4, be sure to raise the ball screw spline shaft as much as possible.<br>If the ball screw spline shaft is lowered, the shaft will be damaged early.<br>- To operate axes 1, 2, and 4 with the ball screw spline shaft lowered, use the SPEED command and ACCEL/DECEL command to adjust the operating speed and acceleration/deceleration to prevent the ball screw spline shaft from vibrating. |

# Installation

This part describes information you should know before unpacking, transporting, installing, and operating the robot.

## 2. Unpacking and Transportation

The worker should receive safety training defined by each country's laws and regulations. Be aware that the failure and accident resulting from the work done by the customer will not be under our warranty.

Recommended protector:

| Type and name | Protection part and use | Recommended example |
|---|---|---|
| Helmet | Protection part: Head<br>Use: Protect from a falling object.<br>Protect from collision with the arm. |  |
| Safety glasses | Protection part: Eyes<br>Use: Protect from a flying object.<br>Protect from collision with the arm. |  |
| Protective gloves | Protection part: Hands and fingers<br>Use: Protect hands and fingers when caught in the machine.<br>Prevent a carried object from dropping. |  |
| Protective shoes | Protection part: Feet and toes<br>Use: Protect from a falling object. |  |

**⚠ CAUTION**

**Required**

- Be sure to convey the product by two or more workers.
  Working in an unnatural posture causes the packing box to drop.
  - Secure the transportation route in advance.
  - Use the cart, etc. to carry the product.
  - Wear the protective shoes.

**NOTICE**

**Required**

- Be sure to convey the product with by two or more workers.
  Working in an unnatural posture causes the packing box to drop and the product in the box to be damaged.
  - Do not use the dropped product.

## 2.1 Unpacking

The robot is shipped in corrugated cardboards. Fig. 2.1 shows the packing state. Open the packages in a location easily accessible, where the equipment is to be installed. Take careful precautions not to damage the robot. After opening the packages, make sure that all the accessories are present and that nothing has been damaged during transport. For robot and controller accessories, see the accessory list packed with the controller.

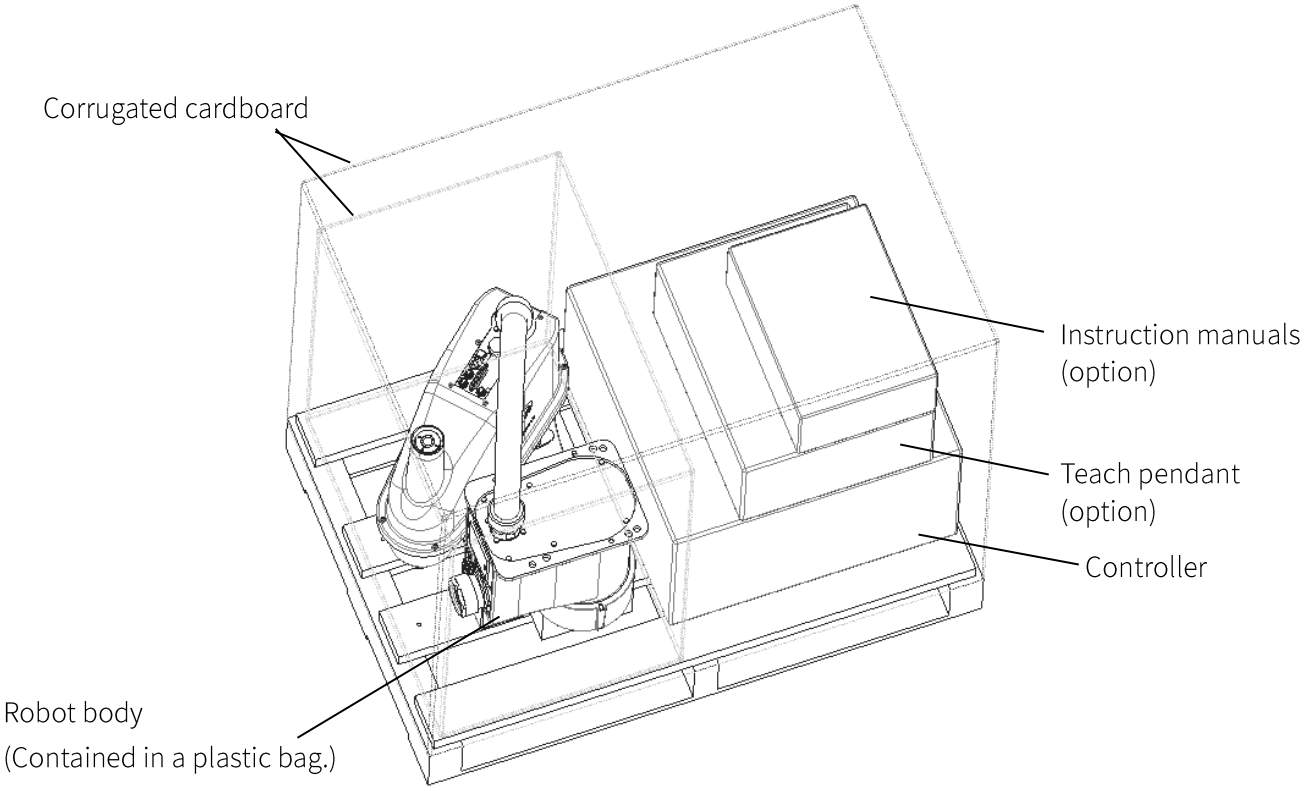



Fig. 2.1 Packaging state（THE600-T）

Table 2.1 Size and Mass

| | |
|---|---|
| Packing size | 630 mm (W) × 900 mm (H) × 920 mm (D) |
| Packing mass | About 66 kg (option not included) |

**⚠ DANGER**

**Caution/ Warning**

- Never use the damaged robot.
  Otherwise, the faulty robot may cause unpredictable disaster.
  - Check the appearance based on the external view described in the manual.
  - Check the parts based on the accessory list attached to the controller.
  - If damage is found or parts are insufficient, please contact SHIBAURA MACHINE Sales Division.

| ⚠ CAUTION | |
|---|---|
|  **Prohibited** | • When cutting the packing rope with the cutter, never direct the blade toward a person.<br>Otherwise, the person may get a cut due to the cutter.<br>- Use the nippers or scissors.<br>- Wear gloves.<br>• During transportation, be sure to always use a forklift to raise the packing box.<br>Otherwise, the packing box could drop, or it could fall on the foot of an operator.<br>- Do not allow people near the transportation route.<br>- When lifting, align the center of the packing box with the center of the fork.<br>• When putting the robot, cables, etc. in and out of the packing box, never lift them up in an unnatural posture.<br>Otherwise, excessive burden may damage worker's back.<br>- Secure space where work can be performed easily.<br>- Put the robot, cables, etc. out of the packing box by multiple workers according to the situation. |

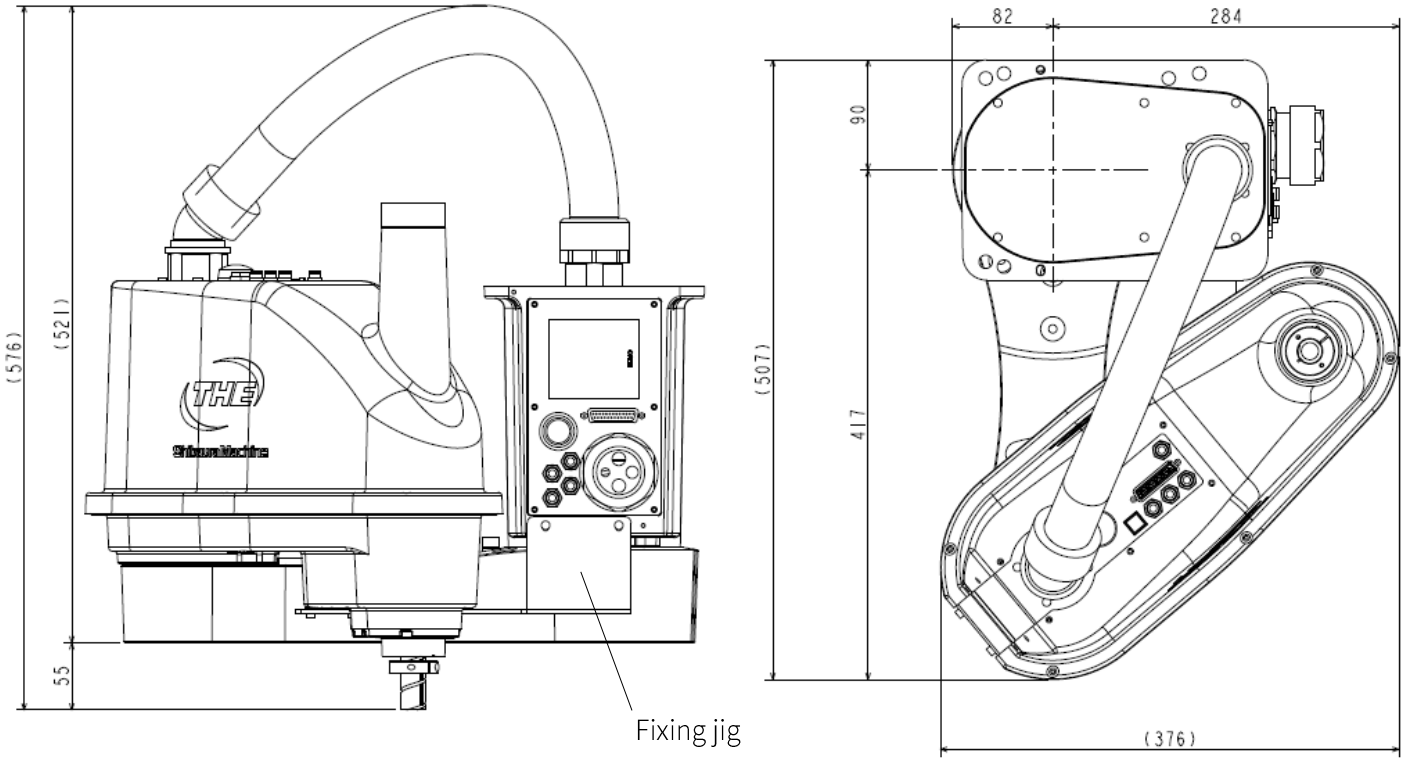
| NOTICE | |
|---|---|
|  **Prohibited** | • Never use the damaged robot.<br>The robot is faulty.<br>- Check the appearance based on the external view described in the manual.<br>- Check parts based on the accessory list attached to the controller.<br>- If damage is found or the parts are insufficient, please contact SHIBAURA MACHINE Sales Division. |
|  **Required** | • Be sure to dispose of the unnecessary packing materials according to the customer's in-house regulations.<br>To store the packing materials, be sure to manage them to prevent them from being used for other than packing the robot. |

## 2.2 Transportation

Move the robot very carefully. Make sure that no excessive impact or vibration is exerted on the equipment. If the equipment is to be subject to vibration over a long period, be sure to tighten all the clamp and base set bolts completely. When transporting the product to a separate location, secure it with the band first as it was packed, and then put the product back into the corrugated cardboards.

## 2.3 Mass and Dimensions

Fig. 2.2 shows the mass and external dimensions of the robot.




Robot weight: 31kg

Fig. 2.2 Outer dimensions at transport (THE600-T)

## 2.4 Transporting the Robot

In principle, the robot should be conveyed in the state shown in Fig. 2.2Outer dimensions at transport (THE600-T)and Fig. 2.3.

Fold back and secure the arm with the attached clamp. (The equipment is shipped in this posture. After you have unpacked the shipment, you should move it as it is.) At this time, be careful not to impose a large force on the spline shaft.

| ⚠ CAUTION | |
|---|---|
| **Prohibited** | • Never convey the robot by one worker.<br>Otherwise, the mass exceeding 31 kg cannot be supported, and the robot will drop.<br>- Convey the robot by two or more workers.<br>- Use the cart, etc.<br>- Secure the conveyance route in advance. |
| **Required** | • When conveying the robot, be sure to use the attached clamp.<br>Otherwise, the worker may get caught in the robot arm that moved.<br>- Wear gloves to prevent slip and protect hands. |

The robot can be lifted up and conveyed.

Mount the two attached eyebolts as shown in Fig. 2.3, hang wire on the eyebolts, and lift up the robot in the two-point suspension slowly.

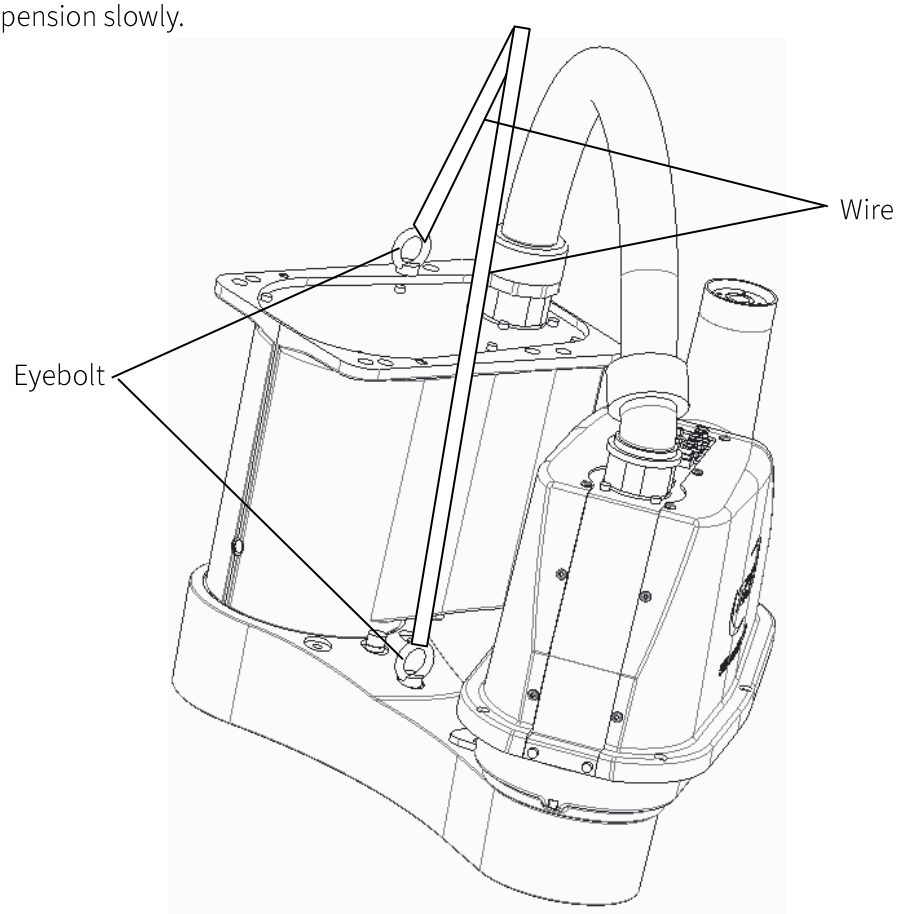
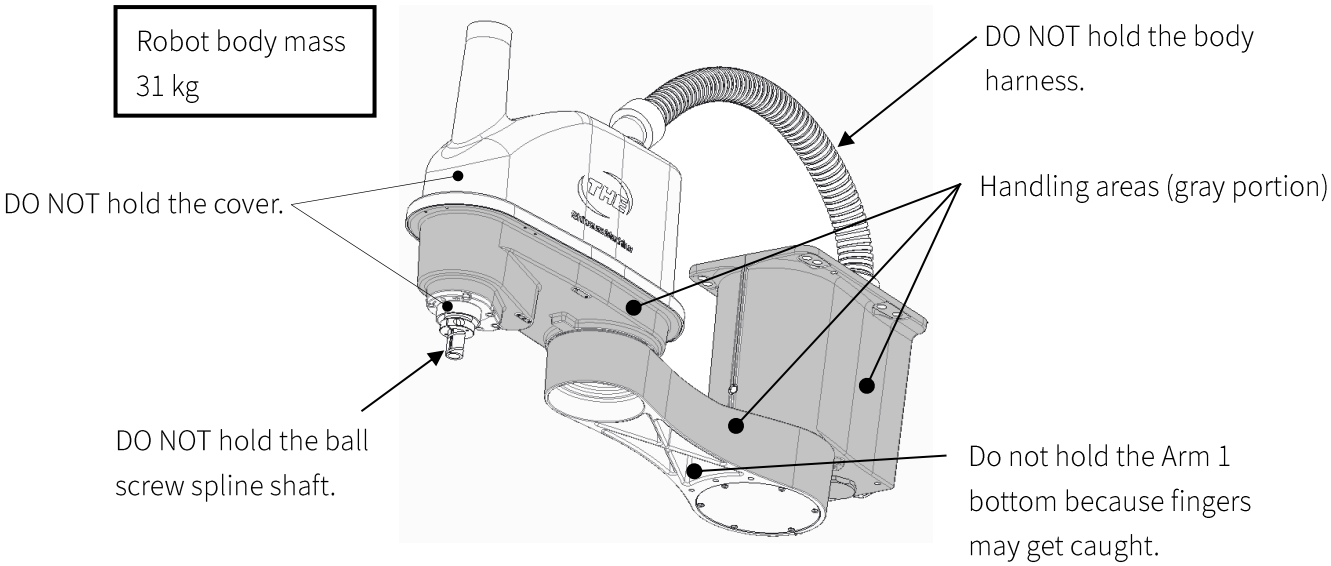



Fig. 2.3 Lifting up the robot

| ⚠ CAUTION | |
|---|---|
|  **Required** | • When lifting up the robot, be sure to lift it up slowly.<br>Otherwise, the tilted robot may collide with the worker.<br>• Use the wire whose limit load can sufficiently bear the weight of the robot.<br>Otherwise, the wire may be broken and the robot may drop. |

| NOTICE | |
|---|---|
|  **Required** | • Be sure to lift up and lift down the robot silently.<br>Otherwise, the robot may collide with the peripheral equipment, frame, etc. to cause failure or damage. |

If lifting up the robot by workers, see Fig. 2.4 to hold the robot by hands.




*MEMO*

*This figure shows the handing areas, and does not show the conveyance posture. For the conveyance posture, see Fig. 2.2 and Fig. 2.3*

Fig. 2.4 Handling areas of the robot (THE600)

| NOTICE | |
|---|---|
| **Prohibited** | • Never touch the ball screw spline shaft by bare hand.<br>Otherwise, rust will develop early.<br>- When touching the ball screw spline shaft, wear gloves.<br>- If the ball screw spline shaft is touched by bare hand, grease up the ball screw spline shaft. |
| **Required** | • When lifting up the robot by workers, be sure to hold the gray portions shown in Fig. 2.4 by hands.<br>If other portions are held, the held portions may be broken. Excessive force may be exerted on the mechanical section, resulting in a malfunction.<br>• After installation is completed, be sure to remove the clamp and eyebolts.<br>If the robot is moved with them mounted, the robot may be broken or become faulty. |

## 2.5 Storage

Avoid storing the robot and controller for long periods of time after unpacking them.
If this is unavoidable, however, strictly observe the following precautions for storage.

### 2.5.1 Storage Precautions for the Robot

| ⚠ CAUTION | |
|---|---|
| **Required** | • When storing the robot, be sure to secure the base completely.<br>Otherwise, the robot will fall over. |

| NOTICE | |
|---|---|
| **Prohibited** | • Never store the robot at a place exposed to direct sunlight and a hot and humid place.<br>Otherwise, the cable duct, timing belt, and resin parts may deteriorate.<br>• DO NOT store the robot at a place where it is exposed to direct rain or water splashes, or at a place containing any toxic gas or liquid. |
| **Required** | • When storing the robot, be sure to seal it in a vinyl bag and put a desiccant in the bag.<br>This prevents rust from developing on and dust from adhering to the ball screw spline unit.<br>- Before storing the robot, apply grease to the ball screw spline shaft.<br>- For the application method, see "7.6.3 Greasing Up the Ball Screw Spline Unit and Applying Corrosion Inhibitor" in the "STA1171: THE600 Robot Manual".<br>- Before the use, apply the grease to the ball screw spline shaft.<br>• Before operating the robot that has been stored, be sure to perform running-in operation.<br>During storage, the grease viscosity increases. If the robot is rapidly operated at a high speed, poor lubrication may cause failure.<br>- Be sure to perform operation at a low speed (about 20% of override) for five (5) minutes.<br>• Before operating the robot that has been stored, be sure to replace the position detector battery. (See "8.3 Replacing Position Detector Battery" in the "STA1171: THE600 Robot Manual".)<br>Before periodic inspection, an alarm will occur.<br>- If no-energization time is long due to storage, the time to keep data using the battery will shorten.<br>• When storing the robot, use the supplied fixtures to secure the arm and base, and then firmly secure the mounting sections. The robot can be unstable and fall down if it is not secured. |

# 3. Installation of the Robot

The worker should receive safety training defined by each country's laws and regulations.
Be aware that the failure and accident resulting from the work done by the customer will not be under our warranty.

Recommended protector:

| Type and name | Protection part and use | Recommended example |
|---|---|---|
| Helmet | Protection part: Head<br>Use: Protect from a falling object.<br>Protect from collision with the arm. |  |
| Safety glasses | Protection part: Eyes<br>Use: Protect from a flying object.<br>Protect from collision with the arm. |  |
| Protective gloves | Protection part: Hands and fingers<br>Use: Protect hands and fingers when caught in the machine.<br>Prevent a carried object from dropping. |  |
| Protective shoes | Protection part: Feet and toes<br>Use: Protect from a falling object. |  |

## 3.1 Installation Environment

Table 3.1 shows the environmental conditions for the location in which the robot are to be installed.

Table 3.1 Environmental conditions for robot

| Item | Specifications |
|---|---|
| Temperature | In operation: 0 to 40°C<br>In storage: -10°C to 50°C |
| Humidity | 20 to 80% (Non-condensing)<br>DO NOT install the equipment where it may be subject to fluids such as water. |
| Altitude | 1000 m or less |
| Vibration | In operation: 0.98 $m/s^2$ or less |
| Dust | No inductive dust should exist.<br>Consult with SHIBAURA MACHINE first if you wish to use the robot and controller in a dusty environment. |
| Gas | No corrosive or combustible gas should exist. |
| IP (Ingress Protection) rating | Equivalent to IEC60529 IP10 (robot side), and equivalent to IP20 (for the controller) |
| Overvoltage category | IEC60664-1 Class III |

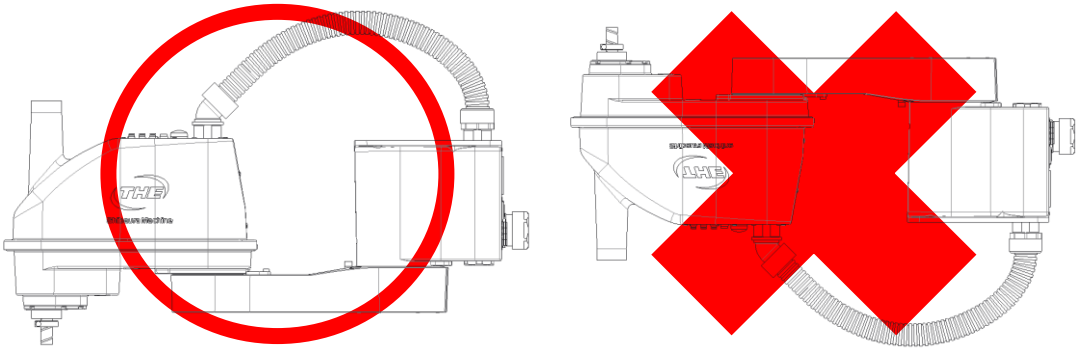
| Item | Specifications |
|---|---|
| Protection against electric shock | IEC61140 Class I |
| Pollution degree | IEC60664-1 Pollution Degree 3 |
| Sunlight | The robot and controller should not be exposed to direct sunlight. |
| Power noise | A heavy noise source should not exist nearby. |
| Magnetic field | A heavy magnetic field source should not exist nearby. |
| Other ambient environmental requirements | No iron powder, no oil, no salt content, and no organic solvent. No water splash on the robot/controller. |

| ⚠ DANGER | |
|---|---|
| **Prohibited** | • Never use the robot at a place where water exceeding the Ingress Protection ratio or liquid other than water splashes.<br>Otherwise, electric leakage may result in electric shock. The electric circuit may be short-circuited to cause fire.<br>• Never place a flammable material near the robot.<br>If the robot ignites due to failure, fire may result. |
| **Caution/ Warning** | • Never use the robot in the high-temperature environment.<br>Otherwise, liquid will easily leak from the alkaline battery used for the position detector battery. If liquid adheres to the skin or an eye, inflammation may result.<br>- When liquid adheres to the skin, wash it away with flowing water immediately.<br>- When liquid adheres to an eye, wash it away with flowing water immediately and then see a doctor.<br>- Replace the parts to which liquid has adhered. |

| NOTICE | |
|---|---|
| **Prohibited** | • Never use the robot at a place where water exceeding the Ingress Protection ratio or liquid other than water splashes.<br>Otherwise, the robot may become faulty. |
| **Required** | • When operating the robot after long-hour stop at a low temperature (10°C or less), be sure to perform operation at a low speed (about 20% of override) for five (5) minutes.<br>The grease viscosity will increase in a low-temperature environment. If the robot is rapidly operated at a high speed, poor lubrication may cause failure.<br>- Before the use, apply grease to the ball screw spline shaft. |

## 3.2 Installation

Before the robot can be installed, the installation layout must be studied in consideration of the working envelope, the coordinate system, and the maintenance clearance.



*MEMO*

*Usage of this robot in an orientation other than the ceiling mount orientation is prohibited.*

**⚠ DANGER**

| | |
|---|---|
| **Required** | • Be sure to connect wiring after the robot is installed.<br>Otherwise, a worker may stumble over the cable and fall. The worker may touch the cable that has been broken due to a dropped carried object and receive electric shock.<br>- House wiring in the duct, etc.<br>• Be sure to connect a grounding wire.<br>Otherwise, the robot that has operated unpredictably due to noise may collide with a worker. Electric leakage may result in electric shock or fire. |

**⚠ CAUTION**

| | |
|---|---|
| **Prohibited** | • When mounting and removing the clamp, never hold the edge of the clamp.<br>Otherwise, if forcible force applies, hand or finger may be cut by the clamp.<br>- Wear gloves. |

| NOTICE | |
|---|---|
| Required | • Be sure to connect wiring after installation is completed.<br>Otherwise, a carried object may drop on the wired cable, resulting in breakage.<br>- House wiring in the duct, etc.<br>• Be sure to connect a grounding wire.<br>Otherwise, the robot may become faulty due to noise. |
| Caution/ Warning | • The cable between the robot and controller is not a movable cable. If the robot is mounted on the traveling shaft, etc., the cable will be broken early. |

### 3.2.1 Installing the Robot

| DANGER | |
|---|---|
| Caution/ Warning | • Never install the robot on a frame that has low rigidity or a weak foundation.<br>Otherwise, the robot may fall over together with the frame during operation.<br>- See the load on the frame during robot operation (Table 3.2) to prepare the frame. |

The robot is secured by using the mounting holes on the base (four (4) places).
Use attached hexagon socket head cap screws M10×25.
Table 3.2 lists the load on the frame during horizontal movement, and Fig エラー! 指定したスタイルは使われていません。.1 shows the robot installation dimensions. The pinholes are provided on the base unit. If the robot position is to be aligned in the base coordinate system or the robot must be replaced, the robot can be positioned by the pinholes.

Table 3.2 Load on the frame during horizontal movement

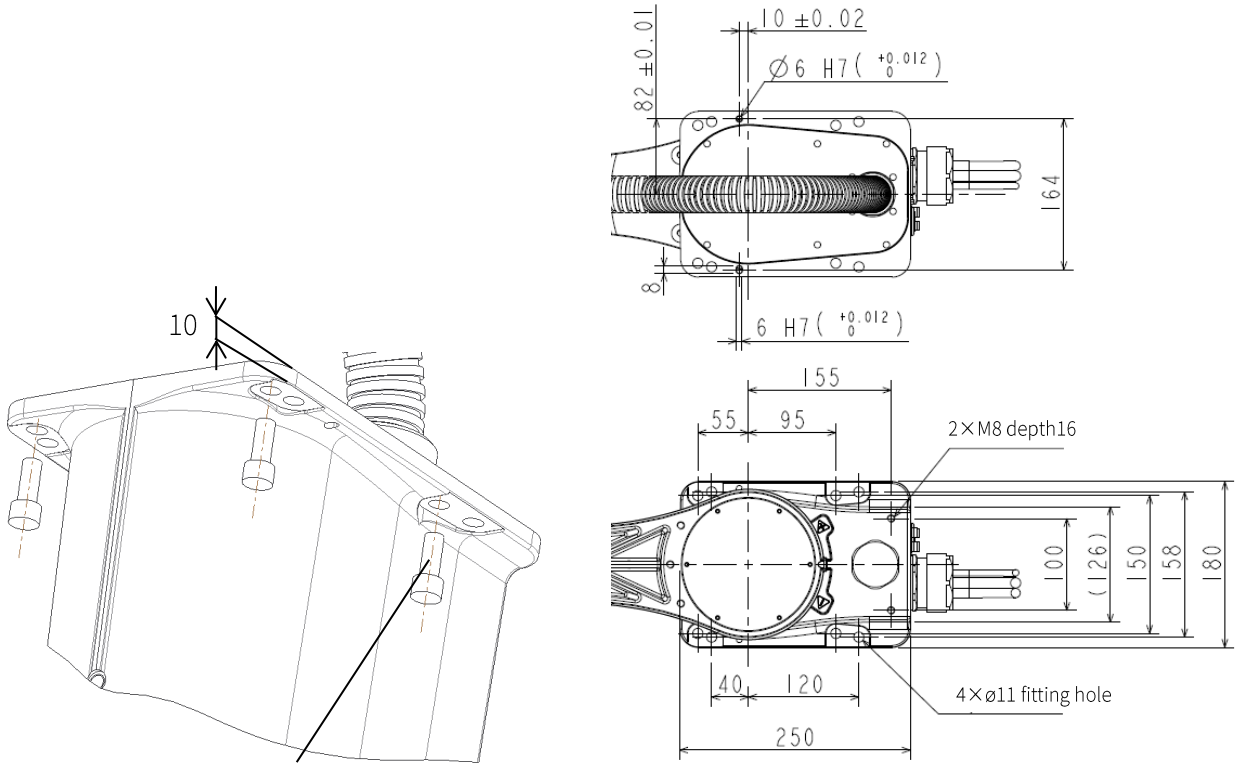
| Model | Load on the frame during horizontal movement (Nm) | Robot body mass (kg) |
|---|---|---|
| THE600-T | 650 | 31 |

*This value is a reference value. When designing the frame, consider safety factor.

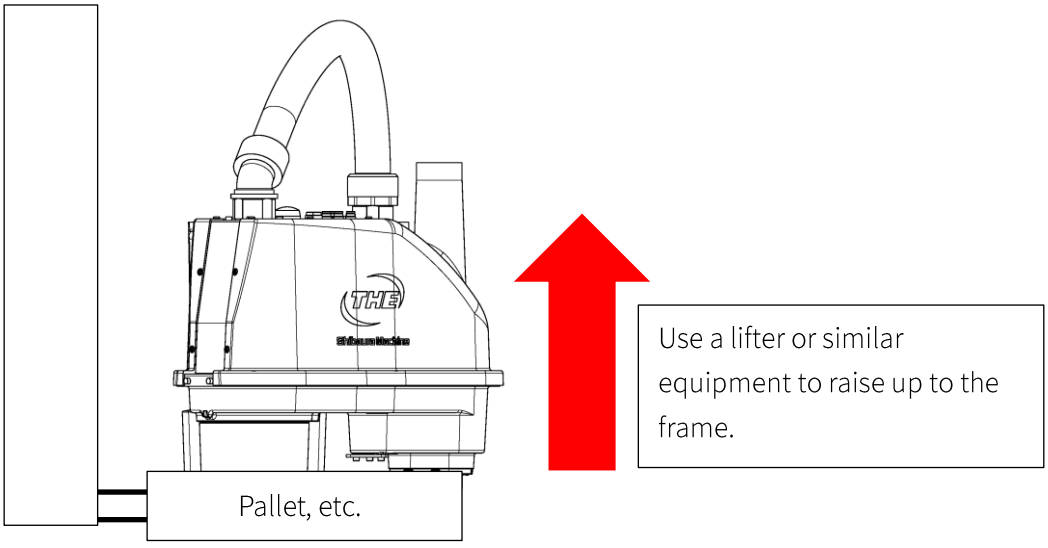
Use the frame that has the flatness and surface roughness listed in Table 3.3 for the frame on which the robot is to be installed.

Table 3.3 Flatness of robot installation part of the frame

| Flatness of robot installation surface | Surface roughness of robot installation surface |
|---|---|
| 0.02 | Ra 6.3 |




Hexagon socket head cap screw M10×25



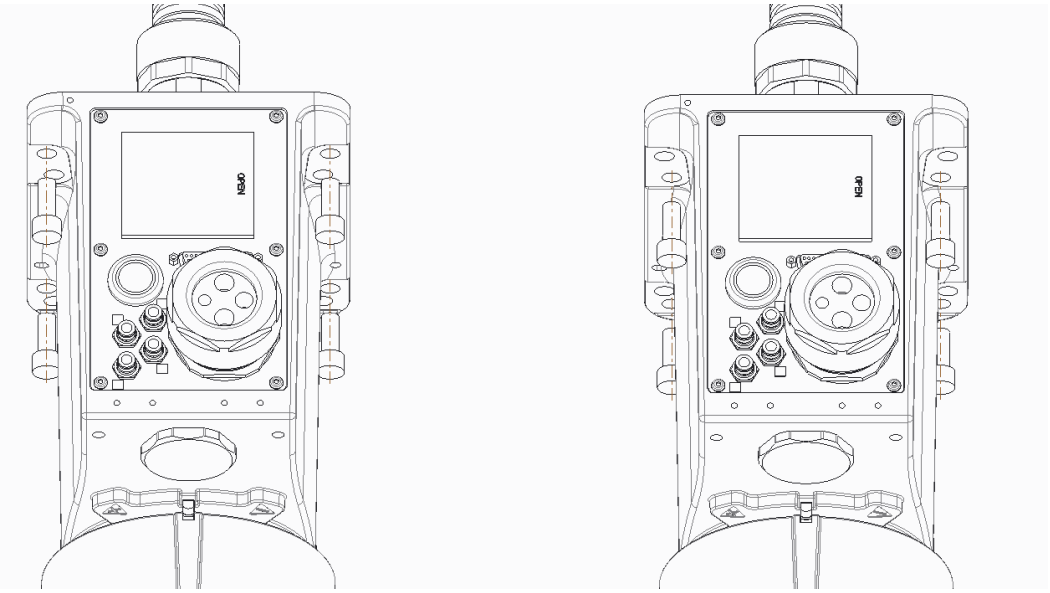

*MEMO*

*Place the robot on a pallet or similar surface, and then use a forklift or similar equipment to transport to the installation frame.*

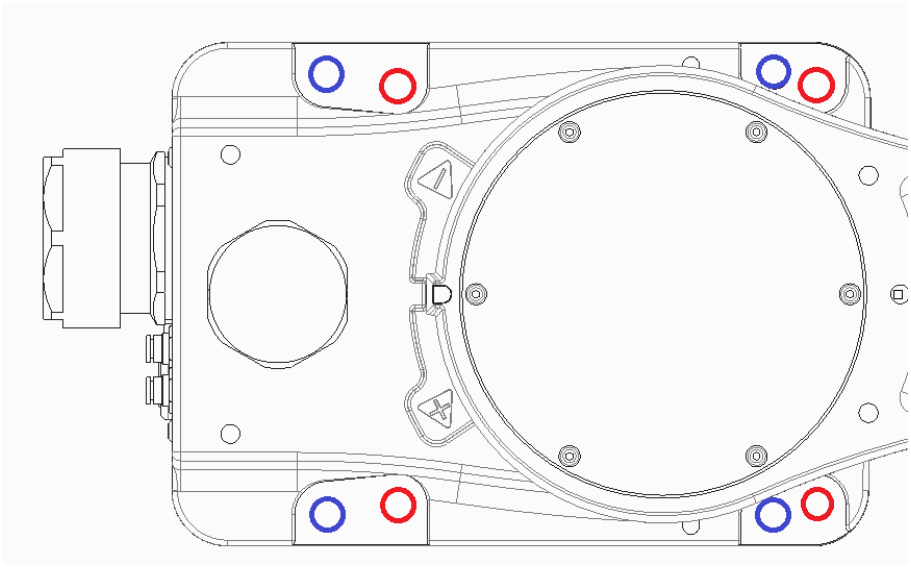
Fig エラー! 指定したスタイルは使われていません。.1 Setting method (THE600-T)

| NOTICE | |
|---|---|
| **Required** | • When placing the robot on the pallet or similar surface, be careful that the ball screw spline does not interfere with the pallet. Otherwise, this could result in breakage or other damage of the ball screw spline. |

The robot base has a total of eight mounting holes for installation. Use four of eight mounting holes to install the robot on the frame. The combination examples of the mounting holes are shown in エラー! 参照元が見つかりません。.



<Example 1> <Example 2>



: Example 1 : Example 2

Fig. 3.2 Combination examples of mounting holes

### 3.2.2 Grounding the Robot

As shown in Fig. 3.3, the robot has the tap hole for grounding. When grounding the robot, use this tap hole to secure the grounding wire terminal with the bolt.

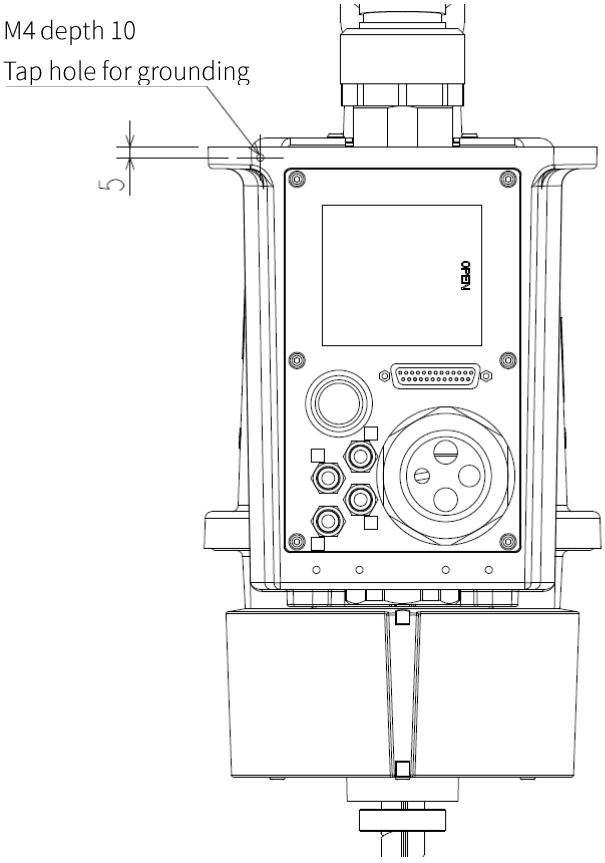



Fig. 3.3 Tap hole for grounding

| NOTICE | |
|---|---|
| **Prohibited** | • Never install the robot on a frame that has low rigidity or a weak foundation. Otherwise, the robot may vibrate during operation. It may become faulty.<br>- See the load on the frame during robot operation (Table 3.2) to prepare the frame. |
| **Required** | • Be sure to install the robot on a level place. Otherwise, the performance and lifespan of the robot may be reduced. The robot may become faulty.<br>• Be sure to secure the frame on which the robot is installed to the floor or wall. Otherwise, the frame will move due to the force exerted during robot operation. |

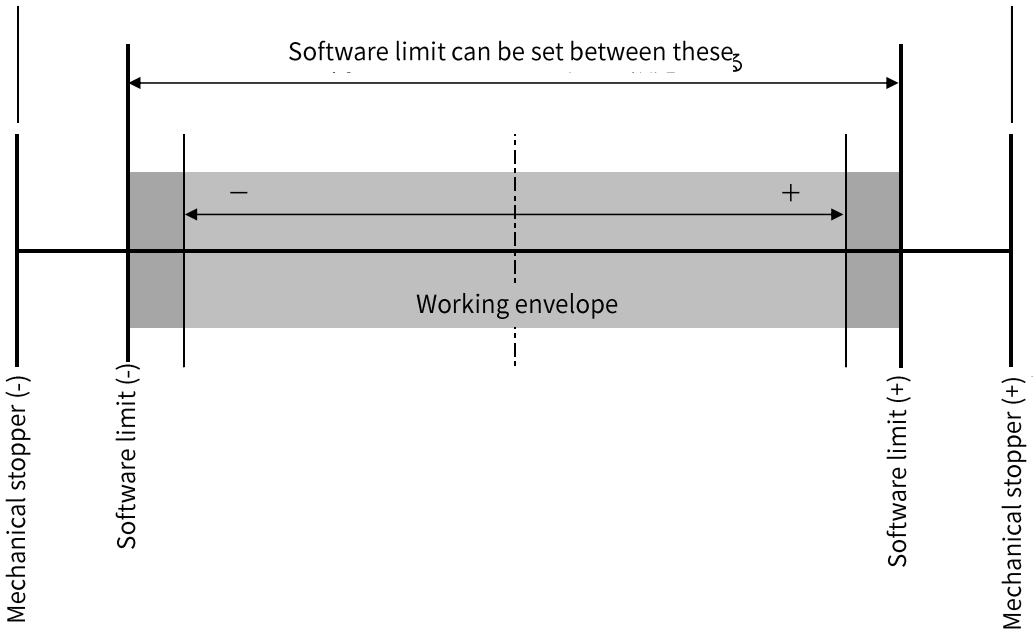
## 3.3 Operating Range

Fig. 3.4 shows the operating range.

Each axis can operate within the working envelope. To prevent the equipment from moving out of the working envelope by misoperation, the robot is equipped with mechanical stoppers outside the working envelope.

Additionally, there are software limits that the user can set up.

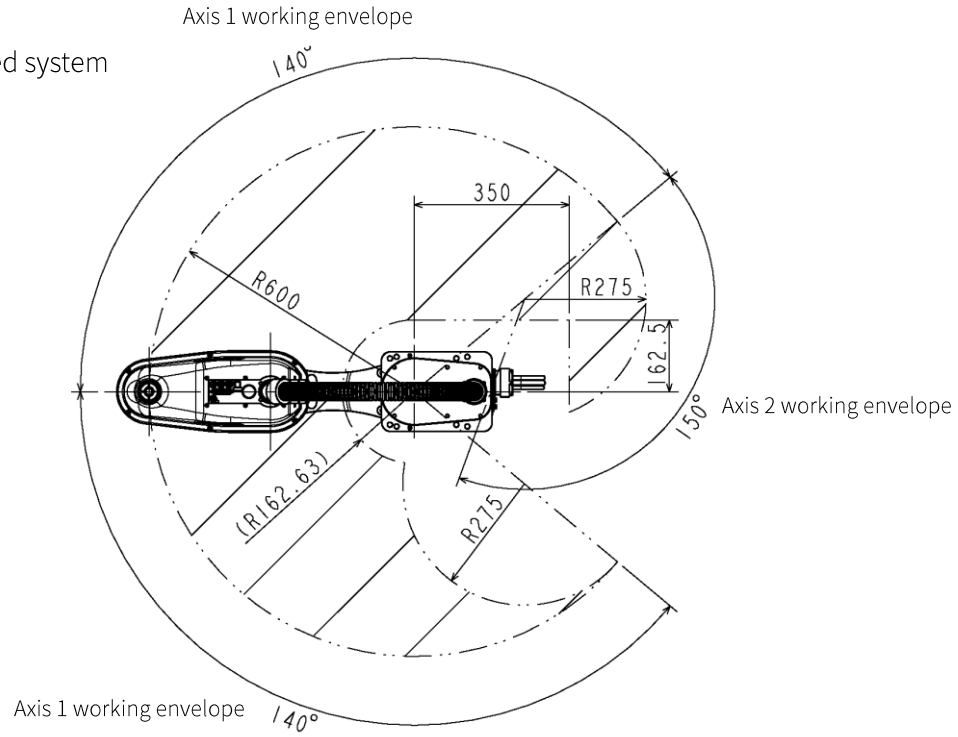
For details, refer to the User Parameters Instruction Manual provided separately.




*MEMO*

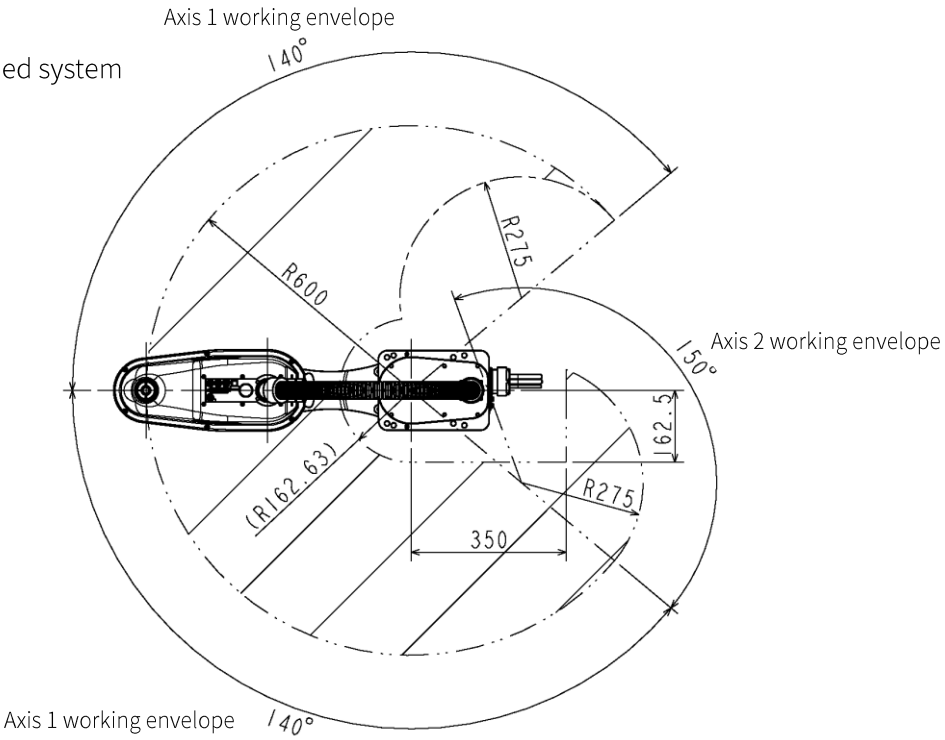
*The software limit cannot be set in the range that is wider than the value set at shipment. The software limit can be set in the range that is narrower than the value set at shipment. So, to change the mechanical stopper position and narrow the working envelope, be sure to change the software limit as well.*

For left-handed system




Left-handed system working envelope

For right-handed system




Right-handed system working envelope

Fig. 3.4 Operating range

### 3.3.1 Changing axis-1 and 2 working envelope

The robot is equipped with the mechanical stoppers to restrict the working envelop of each axis mechanically. Changing the mechanical working envelope of the robot by changing this mechanical stoppers refers to “working envelope change.”
This section explains how to change axis-1 and 2 working envelope of the robot.

For how to change the axis-3 working envelope, see “エラー! 参照元が見つかりません。 エラー! 参照元が見つかりません。” in the “STA1171: THE600 Robot Manual”. Unlike other working axes, the working envelope of axis 4 is restricted by only software limit instead of mechanical stoppers.
Therefore for change of axis-4 working envelope, see “エラー! 参照元が見つかりません。 エラー! 参照元が見つかりません。” in the “STA1171: THE600 Robot Manual”.

| ⚠ DANGER | |
|---|---|
| **Prohibited** | • When turning on the power, never enter the dangerous range of the robot.<br>Otherwise, the robot that has operated unpredictably will collide with a worker while breaking the mechanical stopper.<br>- The mechanical stopper does not aim to stop the robot surely. |
| **Caution/ Warning** | • Before using the robot, be sure to confirm the working envelope.<br>If the working envelope is not limited as intended, the operating robot will collide with a worker. |

| NOTICE | |
|---|---|
|  **Prohibited** | • Never reuse the mechanical stopper that has collided with the robot arm and has been used to change the working envelope.<br>The damaged mechanical stopper will not function normally. |
|  **Required** | • When designing, manufacturing, and mounting the mechanical stopper, be sure to observe the following items.<br>- Be sure to set the changed working envelope to the working envelope indicated in the specification table and external view or narrower working envelope.<br>The robot cannot be operated in the working envelope that exceeds the specifications.<br>- Be sure to re-set zero position information according to “8.9 ③ Other Zero Setting” in the “STA1171: THE600 Robot Manual”.<br>If the mechanical stopper is changed, the function described in “8.9.1.4 How to Restore Data by SYSTEM 1 or SYSTEM2” in the “STA1171: THE600 Robot Manual” becomes disabled. At installation, enter zero data according to “8.9.1.2 Setting HOME1 and HOME2” in the “STA1171: THE600 Robot Manual”.<br>- The function is also available to restore zero using the setting tool prepared by the customer. If using “8.9.1.3 How to Restore Data by HOME1 or HOME2,” enter zero data according to “8.9.1.2 Setting HOME1 and HOME2” at installation.<br>- Breakdowns of the robot due to changing of the mechanical stoppers are outside the scope of the warranty.<br>• Be sure to set the software limit inside the mechanical stopper.<br>Otherwise, the robot will collide with the mechanical stopper during operation. |

The axis-1 working envelope can be changed by changing the position of mechanical stoppers as shown in エラー! 参照元が見つかりません。.

Table 3.4 Axis-1 working envelope before and after change

| | | Before change | After change |
|---|---|---|---|
| Axis-1 working envelope | ＋direction | 140° | 110° |
| | －direction | 140° | 110° |

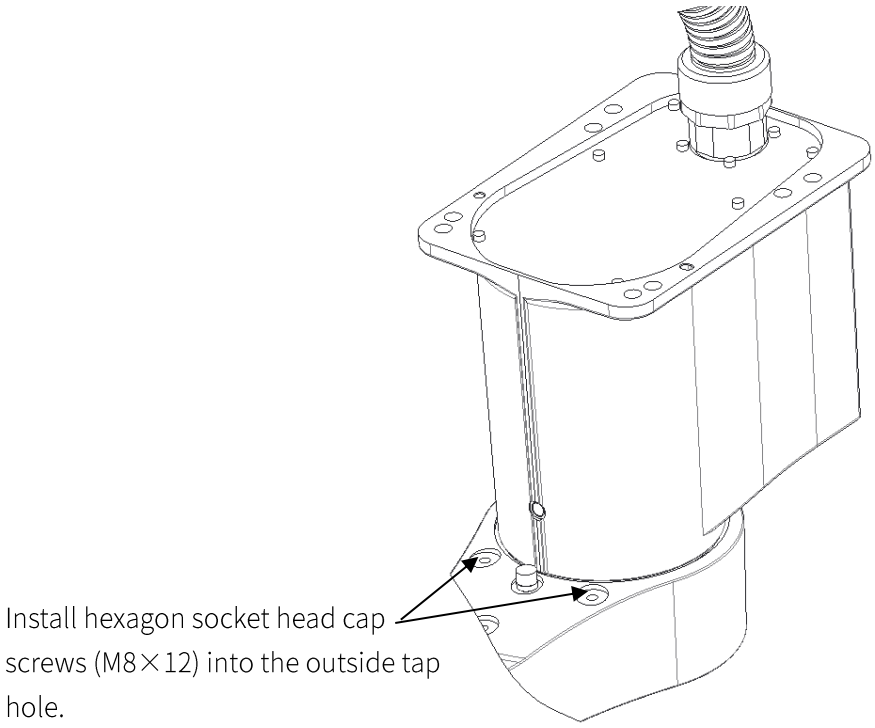



Fig. 3.5 Example of changing axis-1 working envelope

When the axis-2 working envelope is changed, it can be set to any working envelope by changing the shapes of mechanical stoppers. エラー! 参照元が見つかりません。shows the mounting position of the axis-2 mechanical stopper and エラー! 参照元が見つかりません。 and エラー! 参照元が見つかりません。 show the examples of changing the axis-2 mechanical stopper. The customer should design and manufacture the mechanical stoppers as required.

Table 3.5 Axis-2 working envelope before and after change

| | | Before change | After change (Example when changed) |
|---|---|---|---|
| Axis-2 working envelope | ＋direction | 150° | 120° |
| | －direction | 150° | 150° |

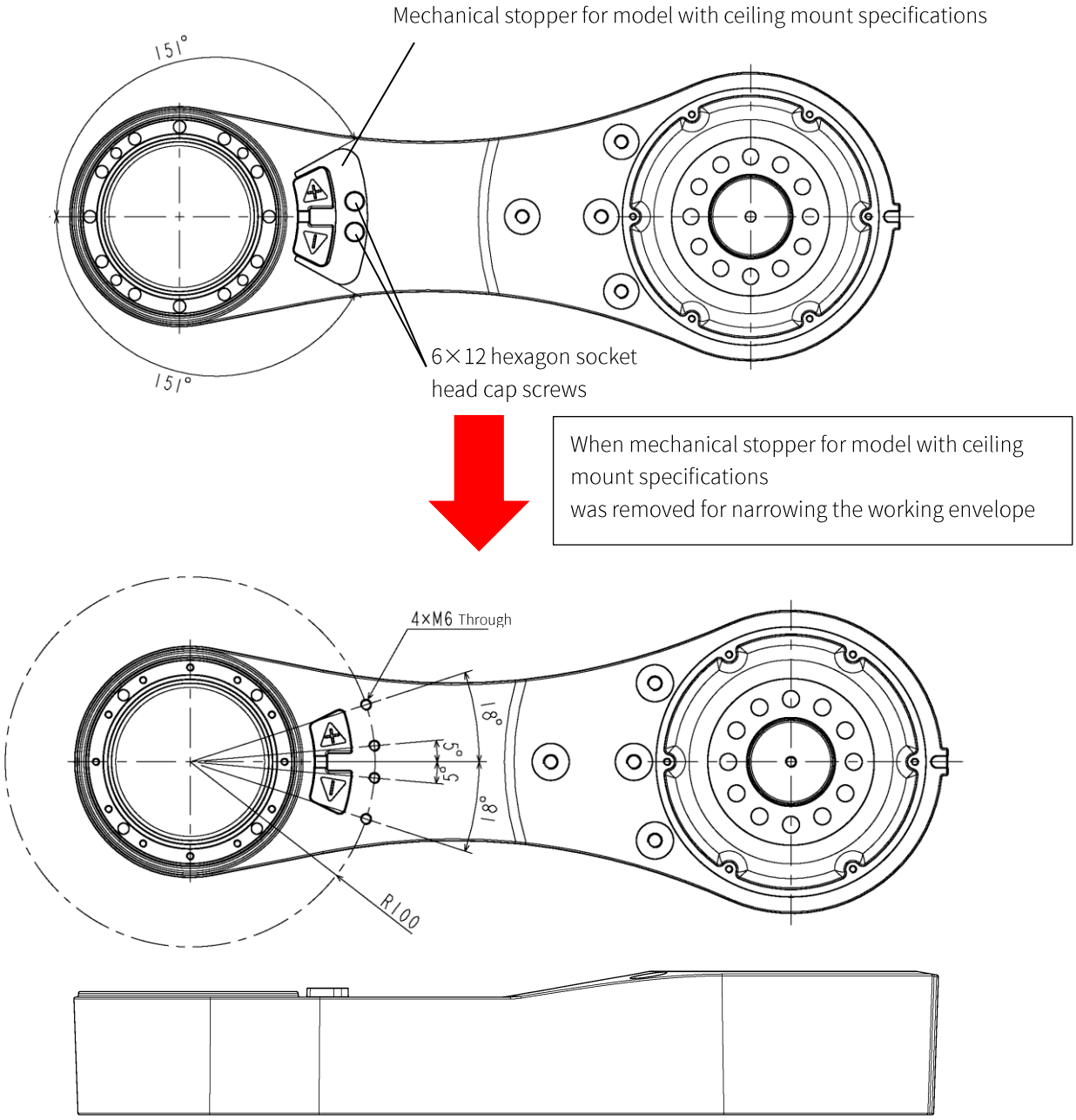



Fig. 3.6 Mounting position of axis-2 mechanical stopper

| NOTICE | |
|---|---|
| **Prohibited** | • Never remove the mechanical stopper for model with ceiling mount specifications except for the purpose of narrowing the working envelope of axis 2.<br>Otherwise, the arm 2 and base will collide, damaging the robot. |

Example of changing axis-2 mechanical stopper
Set working envelope: 120°, -150°

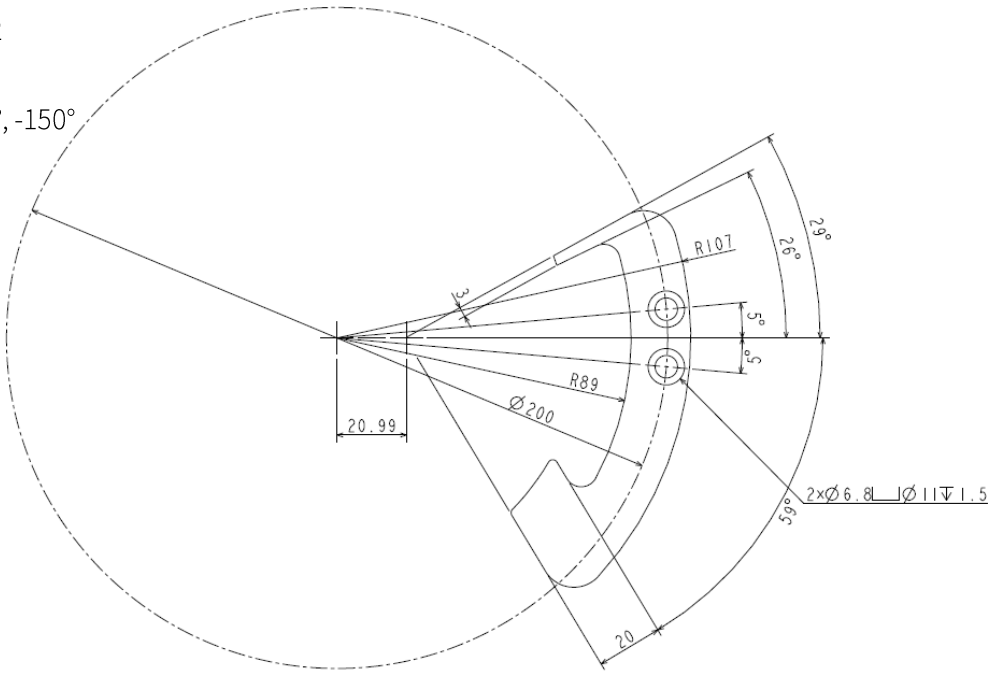



Fig. 3.7 Example of changing axis-2 mechanical stopper

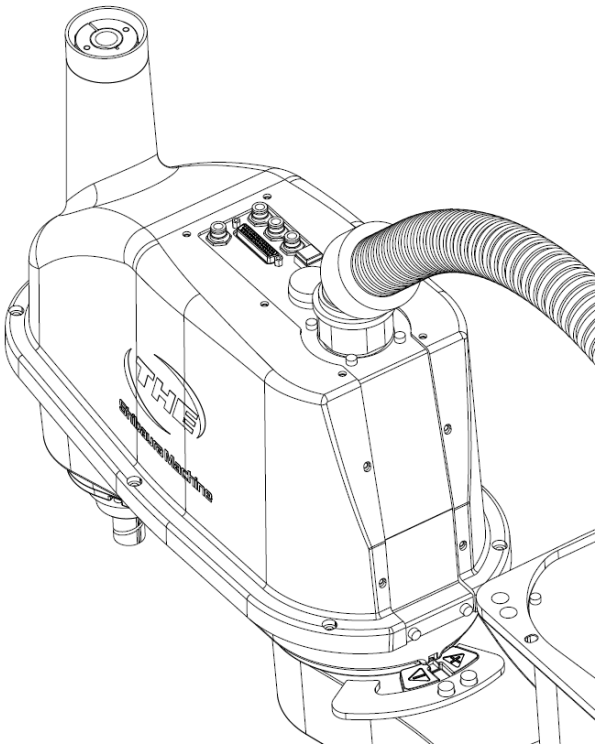


Fig. 3.8 Example of changing axis-2 working envelope

If a working envelope is changed, user parameter must be changed.
For how to change the software limit, see “エラー! 参照元が見つかりません。 エラー! 参照元が見つかりません。” in the “STA1171: THE600 Robot Manual”.

# 4. Setting Hand

The worker should receive safety training for laborer engaged in industrial robot-related work defined by each country's laws and regulations.
Be aware that failure and accident resulting from the work done by the customer will not be under our warranty.

Recommended protector:

| Type and name | Protection part and use | Recommended example |
|---|---|---|
| Helmet | Protection part: Head<br>Use: Protect from a falling object.<br>Protect from collision with the arm. |  |
| Safety glasses | Protection part: Eyes<br>Use: Protect from a flying object.<br>Protect from collision with the arm. |  |
| Protective gloves | Protection part: Hands and fingers<br>Use: Protect hands and fingers when caught in the machine.<br>Prevent a carried object from dropping. |  |
| Protective shoes | Protection part: Feet and toes<br>Use: Protect from a falling object. |  |

## 4.1 Mounting hand

The hand mounting specifications for the THE600-T are identical to the standard model.
For details, see "4.1 User Wiring" in the "STA1171: Robot Manual".

## 4.2 Mounting camera and air valve

The camera and air valve mounting specifications for the THE600-T are identical to the standard model.
For details, see "4.2 User Wiring" in the "STA1171: Robot Manual".

## 4.3 Hand air piping

Four hand piping lines (Φ6×4) are available. The lines are connected on the base rear and the arm-2 top.

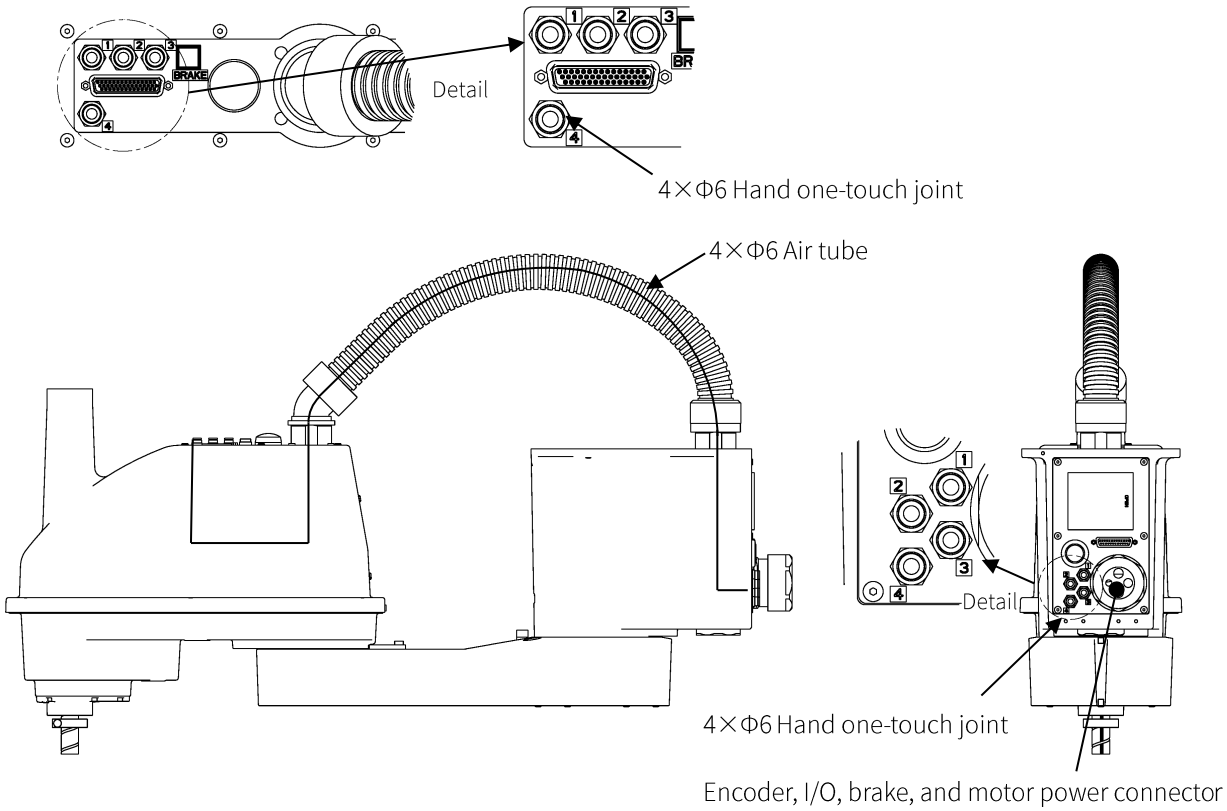



Fig. 4.1 Hand air piping and wiring

| NOTICE | |
|---|---|
| **Prohibited** | • Do not carry out piping together with the body harness.<br>Otherwise, forcible force is imposed on the body harness, which can lead to breakage in the internal cable. |
| **Caution/ Warning** | • The solenoid valve air tubes are not attached to the robot.<br>User should prepare the air tubes. |

## 4.4 Permissible load connections and program setting

The permissible load conditions and program setting specifications for the THE600-T are identical to the standard model.

For details, see “4.4. Permissible Load Conditions and Program Settings” in the “STA1171: Robot Manual”.

# 5. User Wiring

The work should receive safety training defined by each country's laws and regulations.
Be aware that the failure and accident resulting from the work done by the customer will not be under our warranty.

## 5.1 Hand wiring

The hand wiring specifications for the THE600-T are identical to the standard model.
For details, see "5.1 Hand Wiring" in the "STA1171: Robot Manual".

### 5.1.1 Hand control wiring

The hand control wiring specifications for the THE600-T are identical to the standard model.
For details, see "5.1.1 Hand Control Wiring" in the "STA1171: Robot Manual".

### 5.1.2 User wiring

Sixteen (16) signals are provided as user wiring that is controlled by electric gripper, separately placed PLC, etc.
The wiring is connected to the connector on the arm-2 top and on the base rear.
The user should prepare the connector and connect the cable.

Connector (standard) type: <Hood> XM2S-2511 (manufacturer: OMRON)
<Connector> XM3A-2521 (manufacturer: OMRON)
Applicable cross sectional area of wire conductor: AWG 24 to 26 (plain braided wire and twisted wire)
*Connect the connector and wire by soldering.

For the arm-2 side connector, see the connector in "5.1.1 Hand control wiring" in the "STA1171: Robot Manual".

The signal line from the base rear connector "USER" passes inside the robot and is wired to the hand wiring connector "HAND."
Connect the manufactured cable connector to "USER" and "HAND."

*MEMO*

*Hand I/O cable to be connected to the connector "HAND" is also optionally available. For optional hand I/O cable, see "エラー! 参照元が見つかりません。 エラー! 参照元が見つかりません。" in the "STA1171: Robot Manual".*

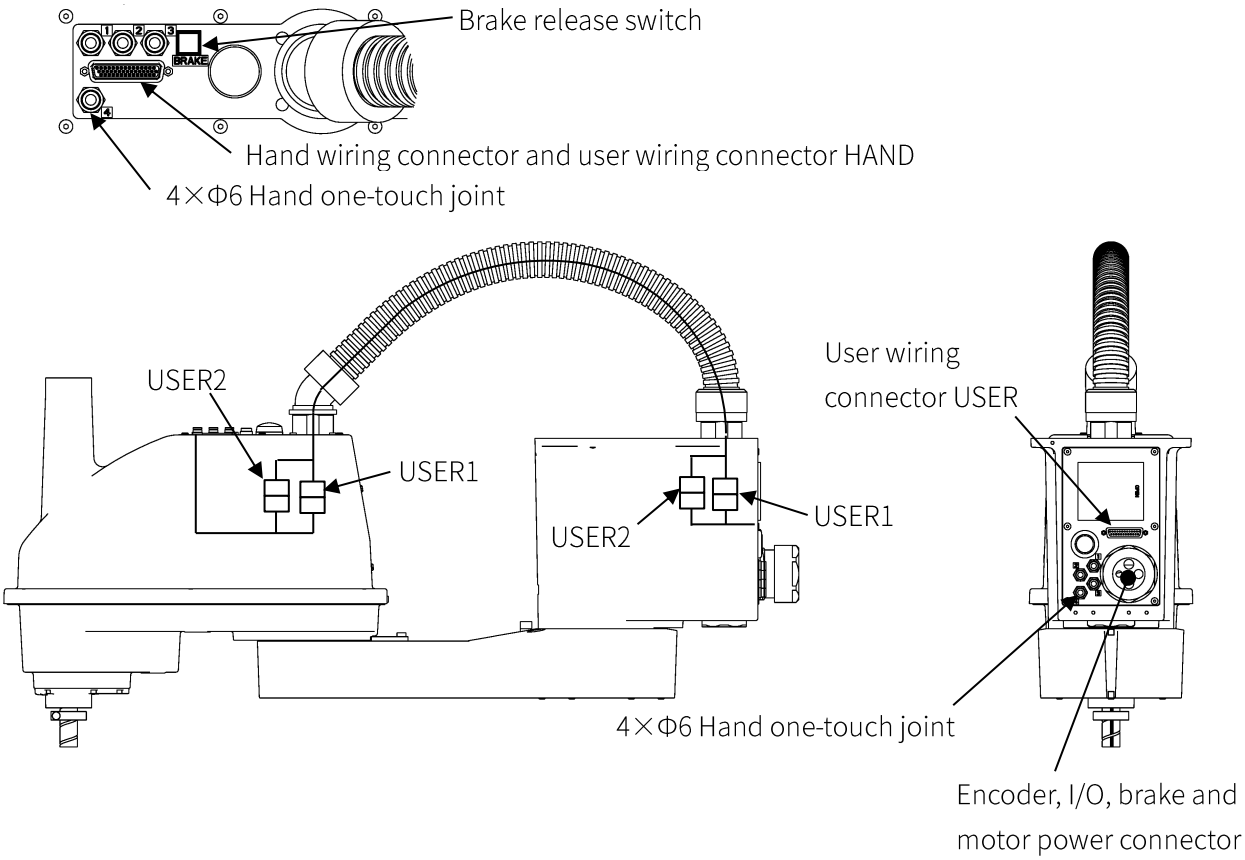



Fig. 5.1 Wiring to PLC

User wiring is as shown in Fig. 5.2.

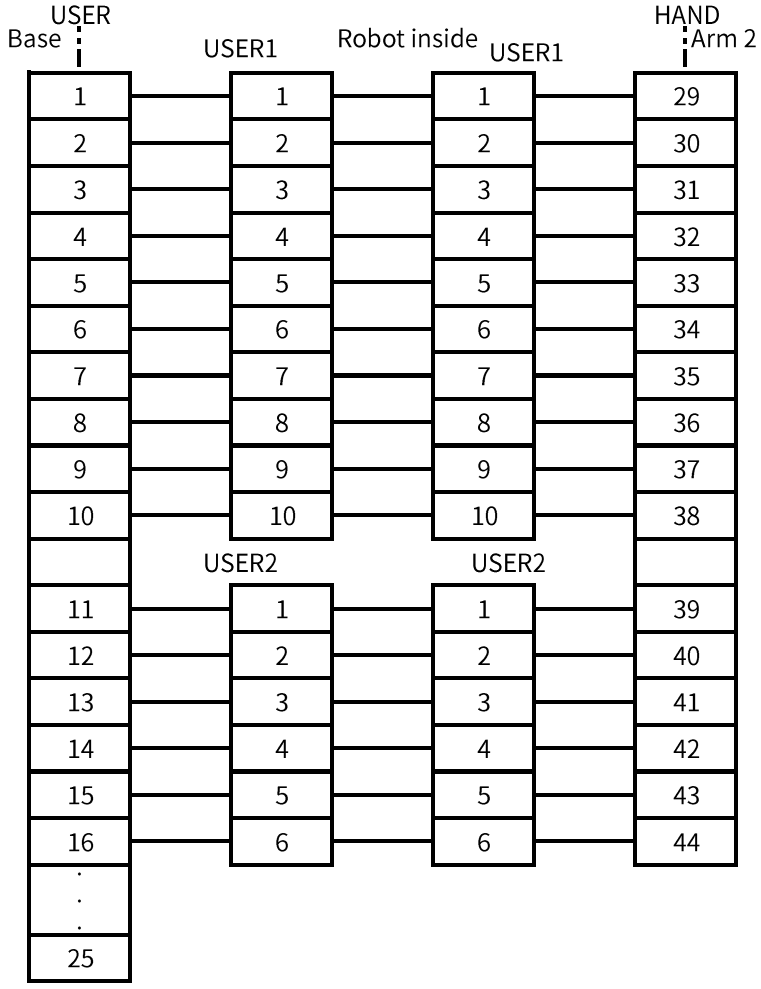



Fig. 5.2 User wiring diagram

## 5.1.3 Specifications of hand connector HAND

The hand connector specifications for the THE600-T are identical to the standard model.
For details, see “5.1.3. Hand Connector Specifications” in the “STA1171: Robot Manual”.

## 5.1.4 Connection example

The hand wiring specifications for the THE600-T are identical to the standard model.
For details, see “5.1.4 Connection Examples” in the “STA1171: Robot Manual”.

# Maintenance Manual

This manual describes how to maintain respective units of the robot.

# 6. Maintenance

The arm mounting direction is different for models with ceiling mount specifications, but the basic structure is identical to the standard model.
The items that differ from the standard model are described below.
For details about the other inspection items, see "MAINTENANCE" in the "STA1171: THE600 Robot Manual".

## 6.1 Configuration of Mechanical Components

Fig. 6.1 shows the configuration of the mechanical components.

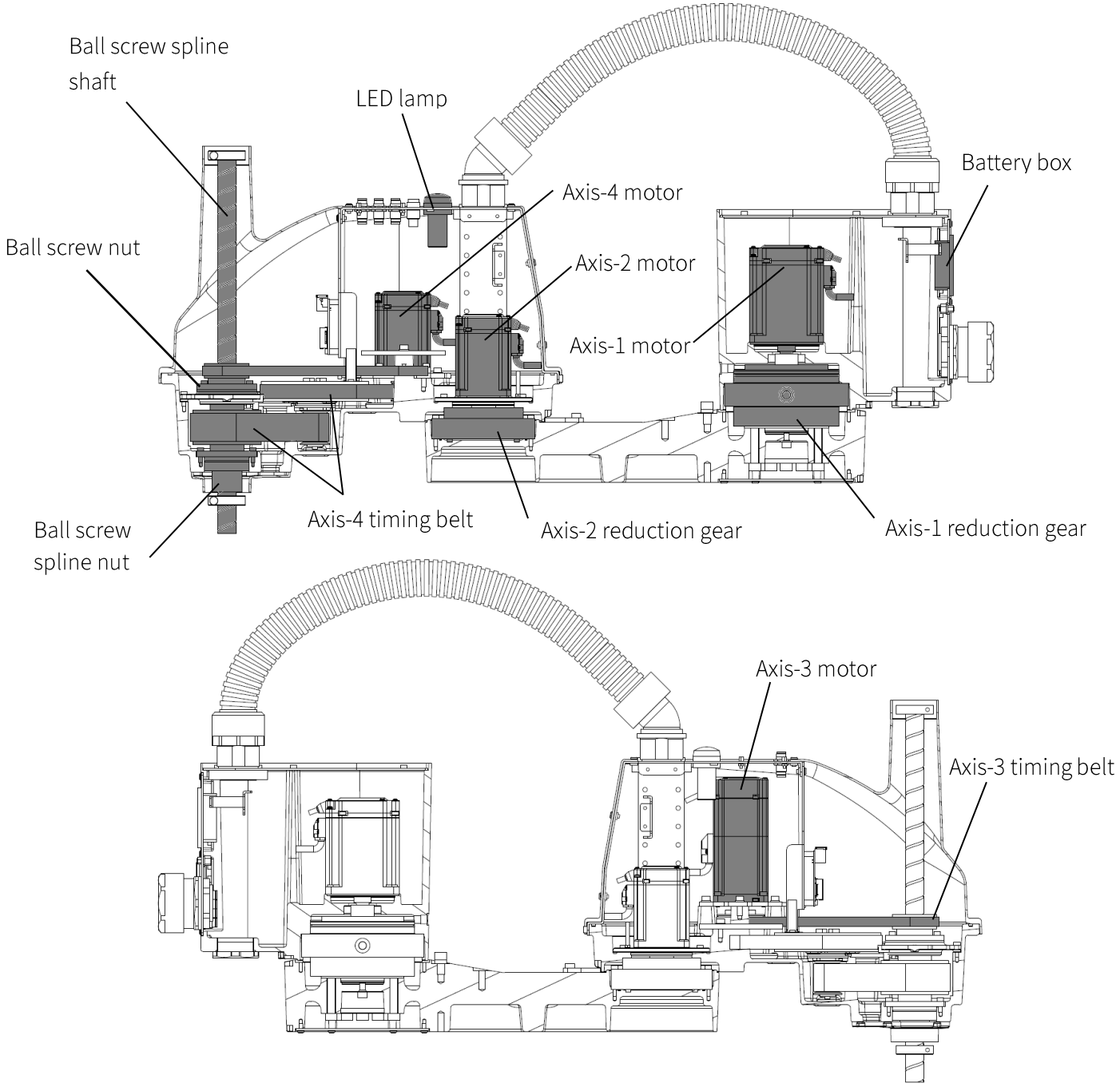



Fig. 6.1 Configuration of mechanical components

### 6.1.1 Arm-1 Cover

The arm-1 cover is secured to the arm-1 lower section using hexagon socket head cap button screws (M4×8×6 screws).

When mounting the arm-1 cover, apply threadlocker adhesive (Recommended: Loctite 425) to the hexagon socket head cap button screws, and tighten at a torque of 1.30 Nm.

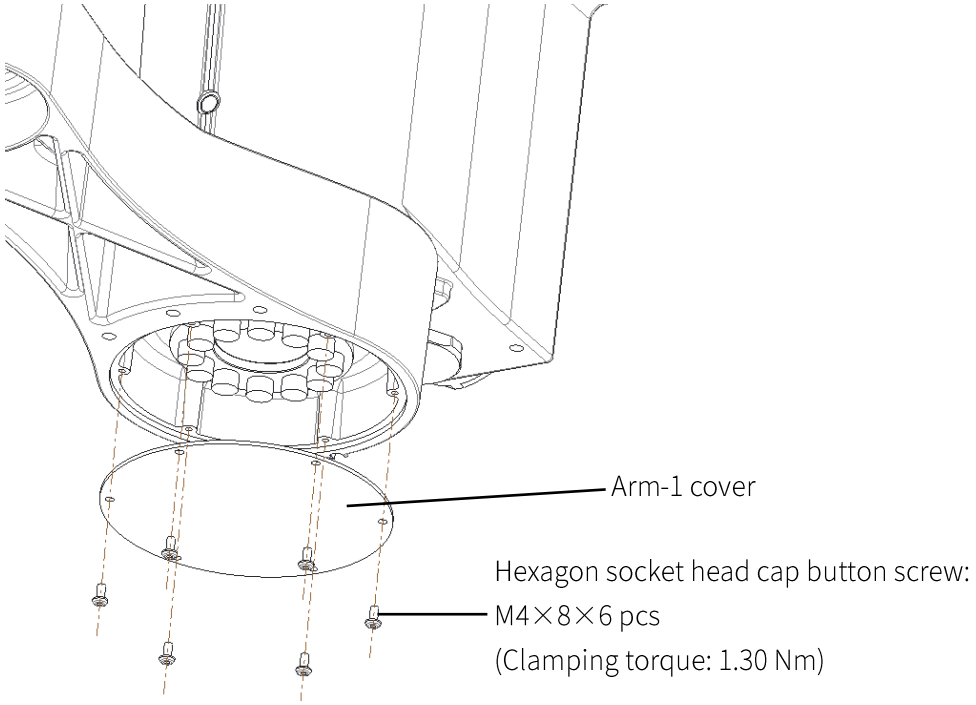
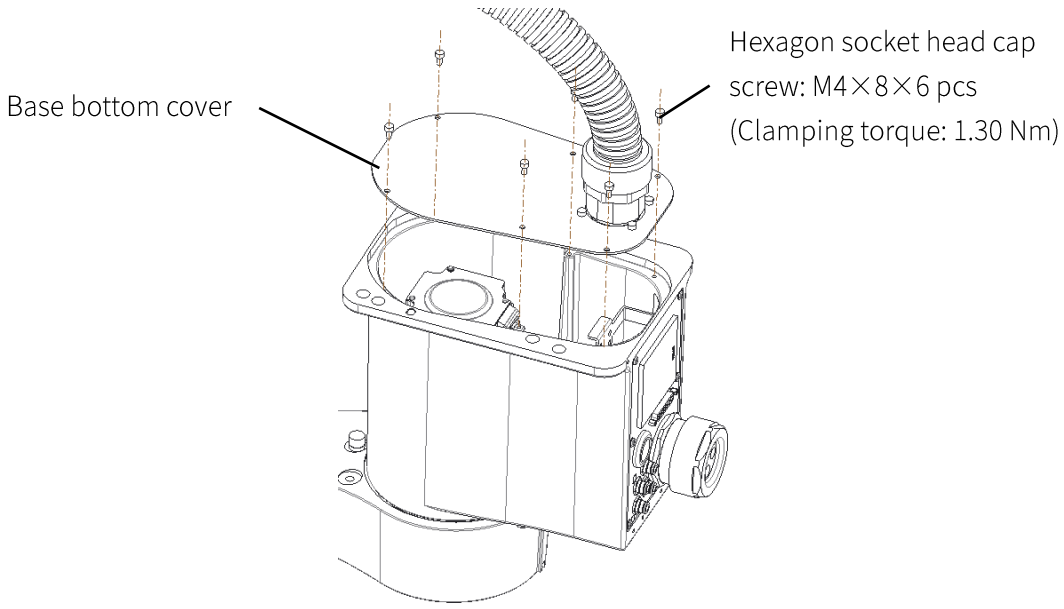



Fig. 6.2 Arm-1 cover

### 6.1.2 Base Cover

The covers for the base include a base rear cover and base bottom cover.
The base rear cover is mounted in the same way as the standard model. For details, see the "STA1171: THE600 Robot Manual".
The base bottom cover is secured to the base using hexagon socket head cap screws (M4×8×6 screws), and the main body harness is connected.
When mounting the base covers, apply threadlocker adhesive (Recommended: Loctite 425) to the hexagon socket head cap screws, and tighten at a torque of 1.30 Nm.




<Base bottom cover>

Fig. 6.3 Base Cover

> *MEMO*
> *Cables are mounted to both the base rear cover and base bottom cover, and so be careful not to pull with excessive force when removing them.*

## 6.2 Replacement of Motors

The worker should receive safety training for laborer engaged in industrial robot-related work defined by each country's laws and regulations. SHIBAURA MACHINE Service Department also accepts replacement work at an extra cost.

Be aware that failure and accident resulting from the work done by the customer will not be under our warranty.

| DANGER | |
|---|---|
| **Required** | • When replacing a motor, be sure to remove the power plug of the controller.<br>Otherwise, the removed cable terminal may be touched to result in an electric shock. The worker may be entangled in rotation of the motor that has made unpredictable motion. |

| NOTICE | |
|---|---|
| **Prohibited** | • Never cause heavy impact on the motor shaft.<br>The motor and position detector may malfunction due to impact.<br>• Never disassemble the motor and position detector.<br>Otherwise, they may be disabled due to difference in position or intrusion of foreign material. |
| **Required** | • When replacing a motor, be sure to remove the power plug of the controller.<br>If replacement work is done with the power turned on, the robot may malfunction.<br>• After the motor is replaced, be sure to re-set zero position.<br>Replacing a motor causes deviation from mechanical zero position; therefore, the robot cannot make normal motion.<br>- For how to re-set the zero position, see "7 Robot Zero Point and Position Detector Error" in the "STA1171: THE600 Robot Manual". |

### 6.2.1 Removing the Axis-1 Motor

**1 Remove the base rear cover and base bottom cover.**

(See "6.1.2 Base Cover.") The rear cover is connected to the connectors inside; therefore, do not pull it forcibly.

**2 Remove the axis-1 motor.**

Remove the hexagon socket head cap screws (M5×16×4 pcs) securing the axis-1 motor, and pull out the axis-1 motor assembly axially. When taking out the motor assembly, do not pull it forcibly as the motor connector is connected to it.

**3 Remove the cable connectors of the axis-1 motor.**

Disconnect cable connectors of the axis-1 motor, J1AS and J1AP and J1BS and J1BP.

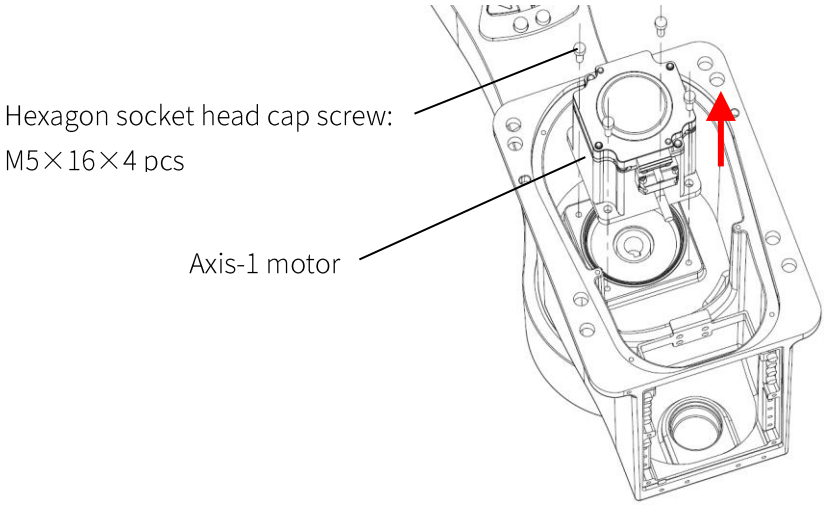



Fig. 6.4 Removing axis-1 motor assembly

**4 Remove the O ring of the axis-1 motor mounting section.**

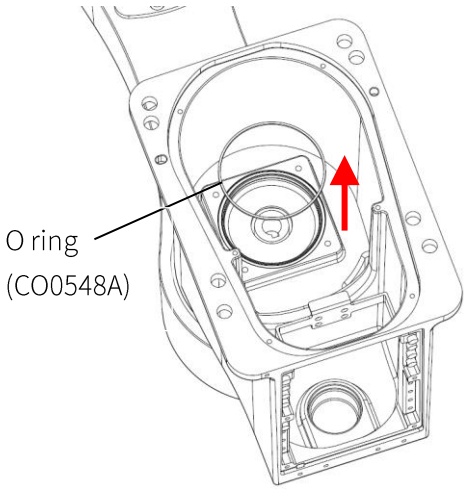



Fig. 6.5 Removing the O ring

### 6.2.2 Fitting the Axis-1 Motor

**1 Mount the O ring to the axis-1 motor mounting section.**

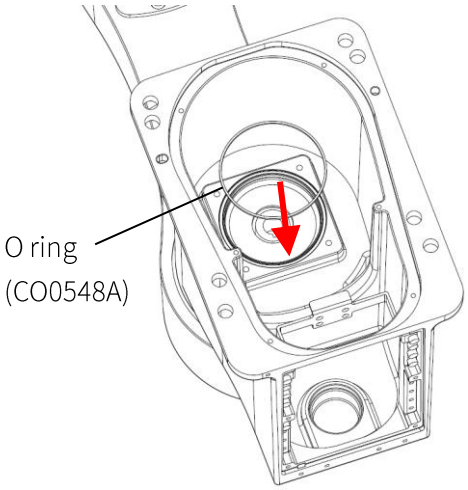



Fig. 6.6 Mounting the O ring

> *MEMO*
> *If the original O ring has become noticeably deformed, obtain and mount a new O ring.*

**2 Insert the axis-1 motor into the base.**

Fit the motor key into the key groove of the reduction gear. At this time, remove the key once, and apply fretting inhibitor to the key side face (see Fig 7.11 (Recommended fretting grease: LGAF 3E)

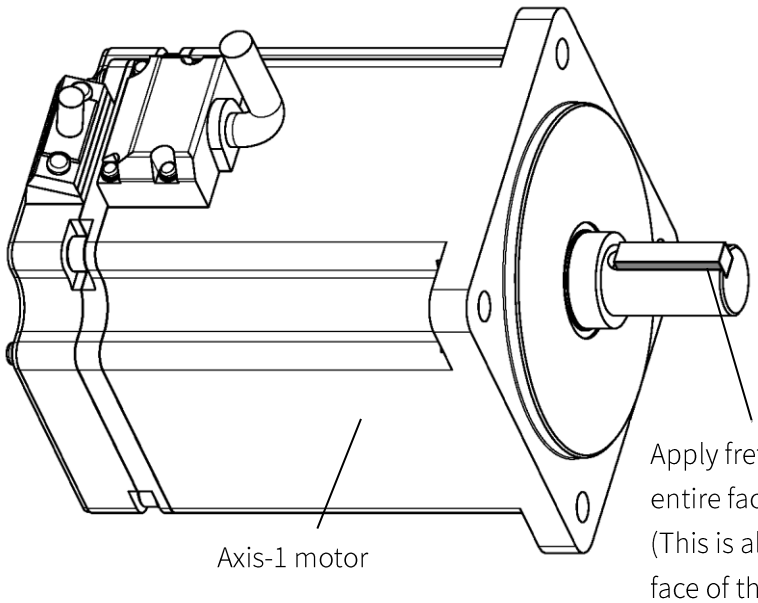



Fig. 6.7 Applying fretting inhibitor

**3 Secure the axis-1 motor assembly with the hexagon socket head cap screws (M5×16×4 pcs).**

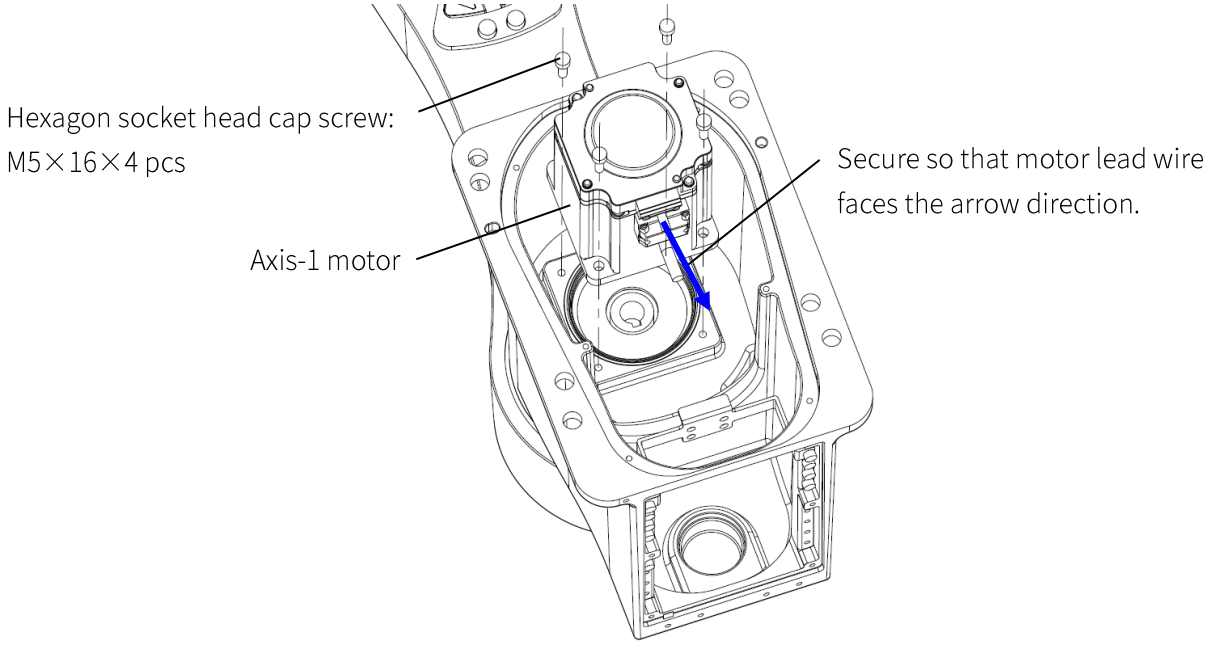



Fig. 6.8 Fitting axis-1 motor assembly

**4 Fit the cable connectors of the axis-1 motor that have been removed.**

Connect the cable connectors of the axis-1 motor, J1AS and J1AP and J1BS and J1BP.

**5 Fit the base rear cover and base bottom cover.**

(See "6.1.2 Base Cover.") After mounting, install the robot in its original location.

**6 Move the arm 1 by hand.**

Make sure that there is no abnormal sound.

**7 Turn on the power and set the zero position of the axis 1.**

The replacement of the axis-1 motor is completed. (Check the coordinates of axes 2, 3, and 4 and set zero position as required. See the "STA1171: THE600 Robot Manual".)

**8 Perform operation at a low speed.**

Perform running-in operation at 20% of override for about one hour to check whether there is any problem with operation of each part.

## 6.3 Replacing Reduction Gears

The worker should receive safety training for laborer engaged in industrial robot-related work defined by each country's laws and regulations. SHIBAURA MACHINE Service Department also accepts replacement work at an extra cost.

Be aware that SHIBAURA MACHINE's warranty does not cover failures and accidents resulting from the replacement of the parts by the customer.

**⚠DANGER**

| | |
|---|---|
|  **Required** | • When replacing the reduction gear, be sure to remove the power plug of the controller. The removed cable terminal may be touched to result in an electric shock. A worker may be entangled in rotation of the motor that has operated unpredictably. |

**NOTICE**

| | |
|---|---|
|  **Required** | • When replacing the reduction gear, be sure to remove the power plug of the controller. If replacement work is done with the power turned on, the robot may malfunction.<br>• After the reduction gear is replaced, be sure to re-set the zero position. Because a mechanical zero position may be shifted with replacement, the robot cannot perform normal operation.<br>- For how to re-set the zero position, see "7 Robot Zero Point and Position Detector Error" in the "STA1171: THE600 Robot Manual". |

## 6.3.1 Removing the Axis-1 Reduciton Gear

**1 Remove the base rear cover and base bottom cover.**

For removal method, see "6.1.2 Base Cover."

The base rear cover is connected with the connectors inside.

Therefore, do not pull it forcibly.

**2 Remove the axis-1 motor.**

For the removal method, see "6.2.1 Removing the Axis-1 Motor."

**3 Remove the arm-1 cover.**

The arm-1 cover is secured with the hexagon socket head cap screws (M4×8×6 pcs).

For the removal method, see "6.1.1 Arm-1 Cover."

**4 Remove the arm 1.**

The arm 1 is secured with the hexagon socket head cap (M8×50×12 pcs).

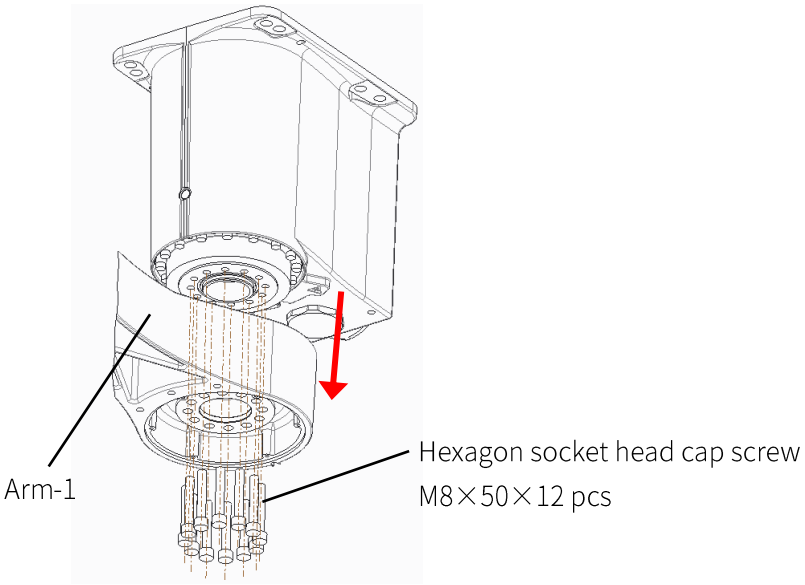



Fig. 6.9 Removing the arm 1

| ⚠ CAUTION | |
|---|---|
| Required | • When fitting or removing the arm, one worker should loosen the clamp bolts and other worker should support the arm.<br>If work is done by one worker, the arm cannot be supported and may drop. |

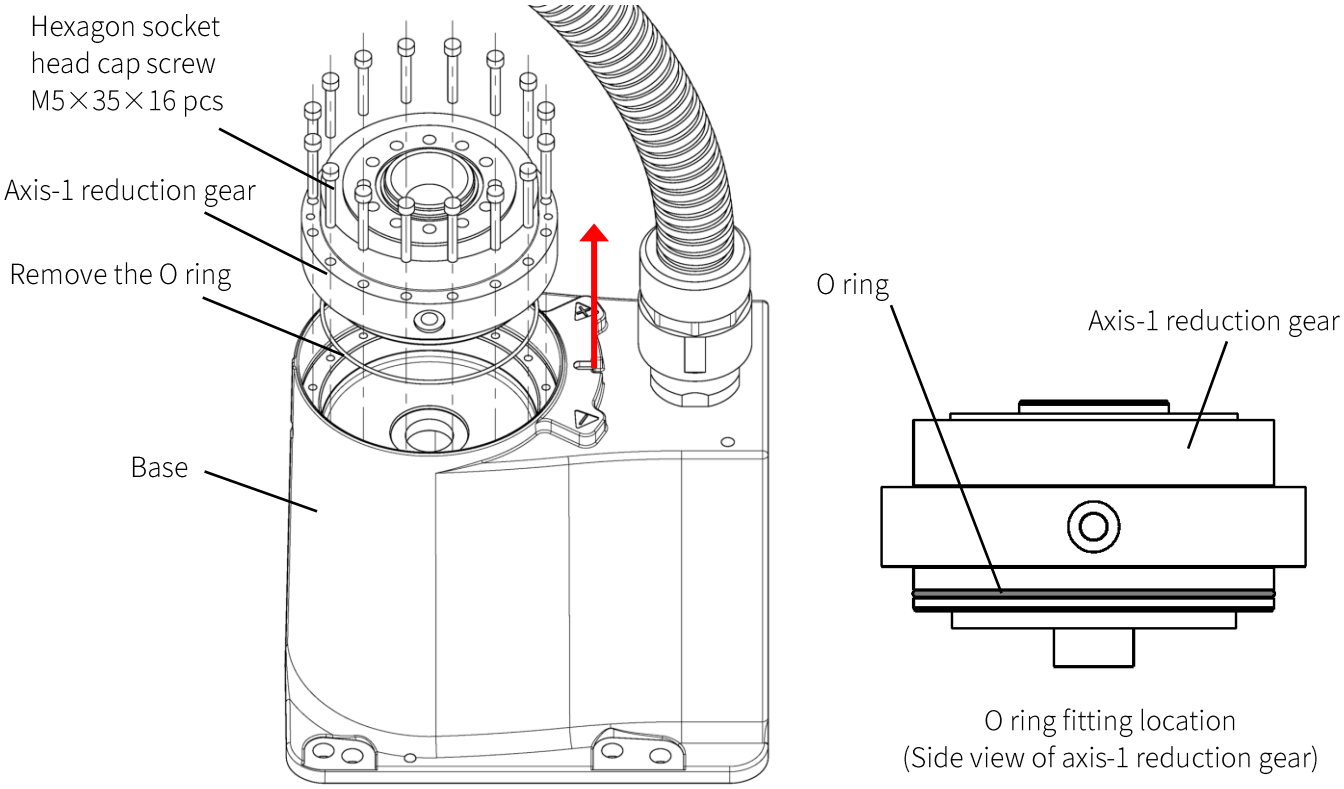
**NOTICE**

**Required**

- Be sure to handle the arm carefully.
  Drop impact or application of excessive external force could ruin the arm.

### 5 Remove the axis-1 reduction gear.

The base and the axis-1 reduction gear are secured with the hexagon socket head cap screws (M5×35×16 pcs).




*MEMO*

*The figure shows the model with standard specifications, but the procedure for mounting and removing the reduction gear is identical to the model with ceiling mount specifications.*

Fig. 6.10 Removing the axis-1 reduction gear

### 6.3.2 Fitting the Axis-1 Reduction Gear

| NOTICE | |
|---|---|
| **Prohibited** | • Never forget fitting the O ring, and never use the cut or damaged O ring.<br>Otherwise, grease will leak from the fitting surface.<br>- Use the new O ring provided for the reduction gear. |
| **Required** | • Be sure to handle the reduction gear carefully.<br>Drop impact or application of excessive external force could ruin the reduction gear.<br>• Be sure to use elliptical cam provided for new reduction gear to be replaced.<br>The combination of the reduction gear and elliptical cam is determined. If they are used in a wrong combination, abnormal sound will occur. The positioning accuracy and lifespan will be reduced. |

**1 Wipe off dust, dirt, remaining grease, etc. well.**

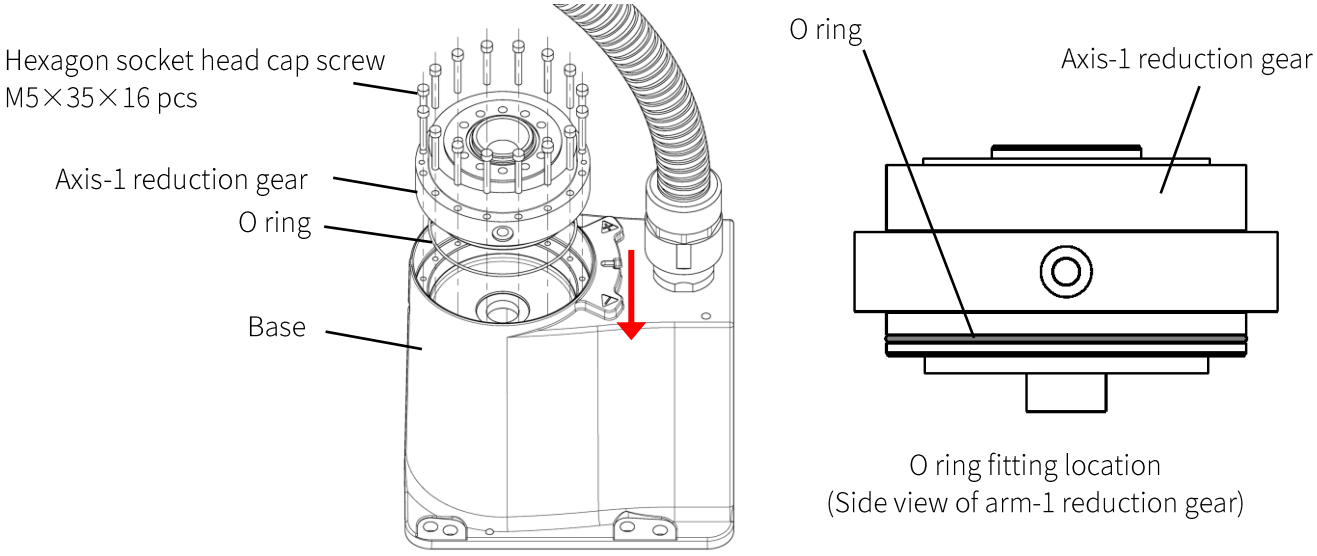
Wipe off the fitting portion of the axis-1 reduction gear in the arm 1 and the O ring groove of the new reduction gear.

**2 Fit the O ring to the axis-1 reduction gear.**

Apply grease to the O ring provided for the new reduction gear, and fit it into the O ring groove.

**3 Fit the arm-1 reduction gear.**

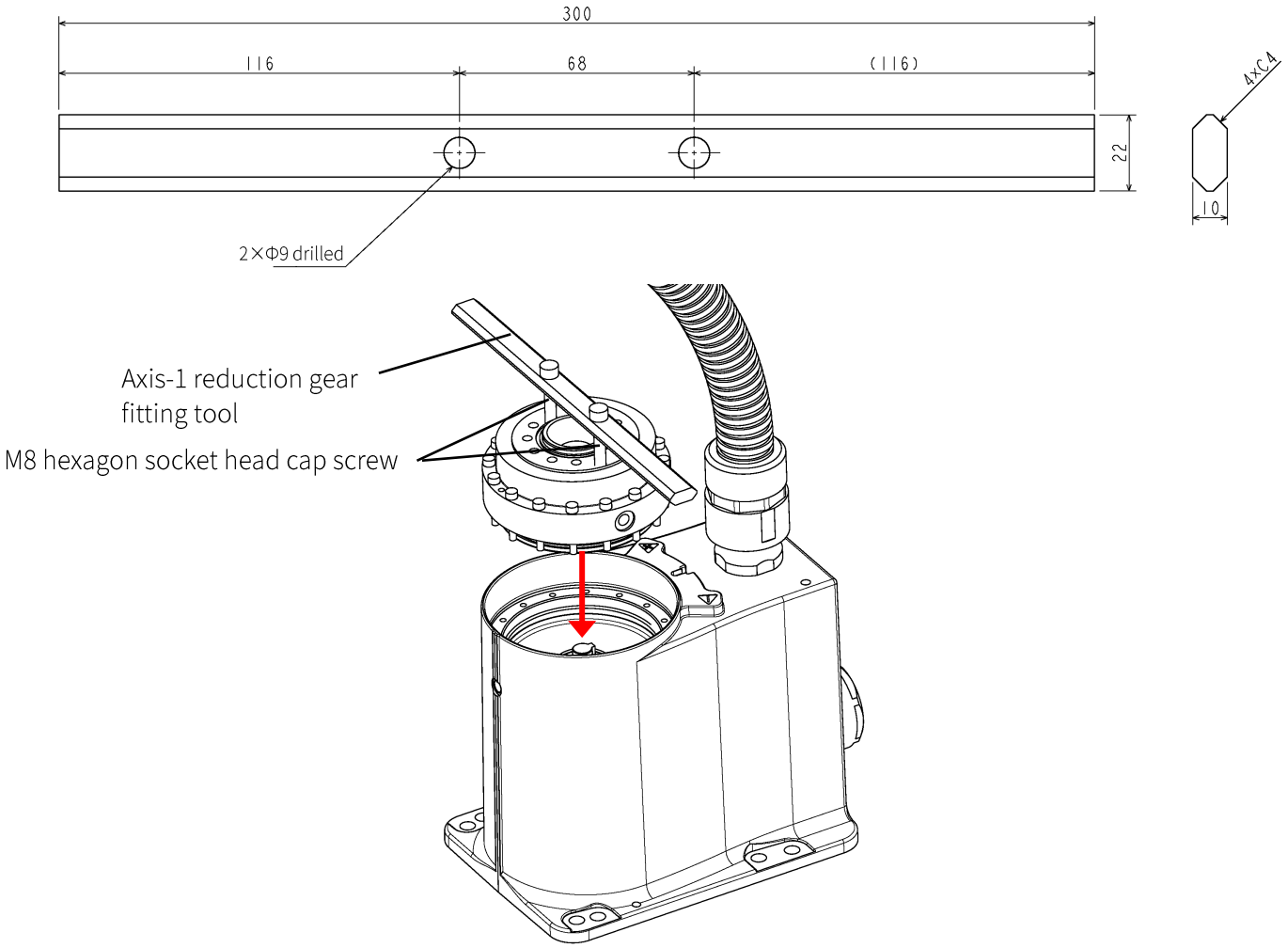
Fit the arm-1 reduction gear with the hexagon socket head cap screws (M5×35×16 pcs). When fitting it, create the tool shown in エラー! 参照元が見つかりません。, and fit it.




*MEMO*

*The figure shows the model with standard specifications, but the procedure for mounting and removing the reduction gear is identical to the model with ceiling mount specifications.*

Fig. 6.11 Fitting the axis-1 reduction gear




*MEMO*

*The figure shows the model with standard specifications, but the procedure for mounting and removing the reduction gear is identical to the model with ceiling mount specifications.*

Fig. 6.12 Example of creating axis-1 reduction gear fitting tool

**4 Fit the axis-1 motor to the axis-1 reduction gear.**

For the fitting method, see "6.2.2 Fitting the Axis-1 Motor."

**5 Fit the base rear cover and base bottom cover to the base.**

For the fitting method, see "6.1.2 Base Cover."

**6 Fit the arm 1 to the arm-1 reduction gear.**

Fit the arm 1 with the hexagon socket head cap screws (M8×50×12 pcs).

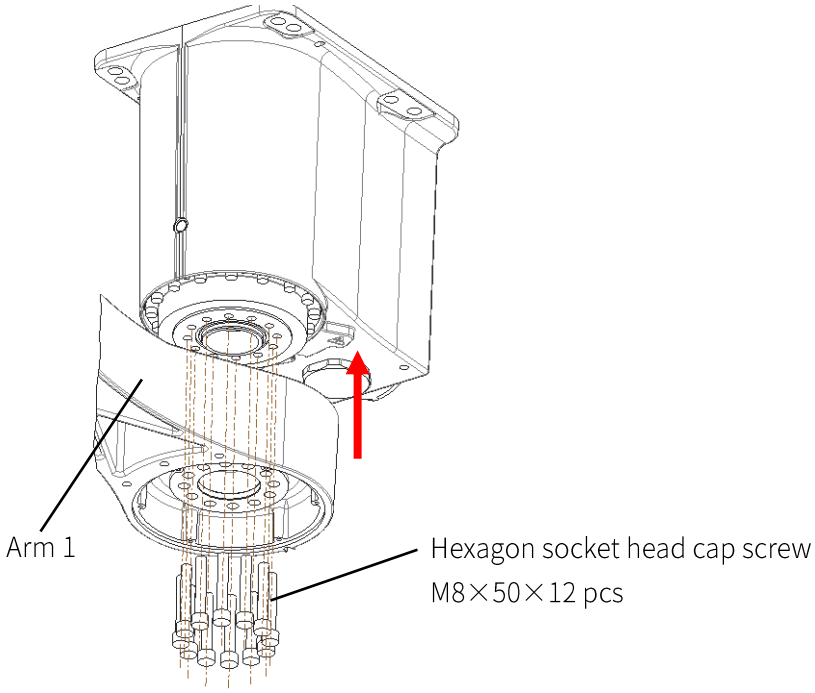



Fig. 6.13 Fitting the arm 1

**7 Remove the hexagon socket head cap screw and seal washer in the arm-1 lower section.**

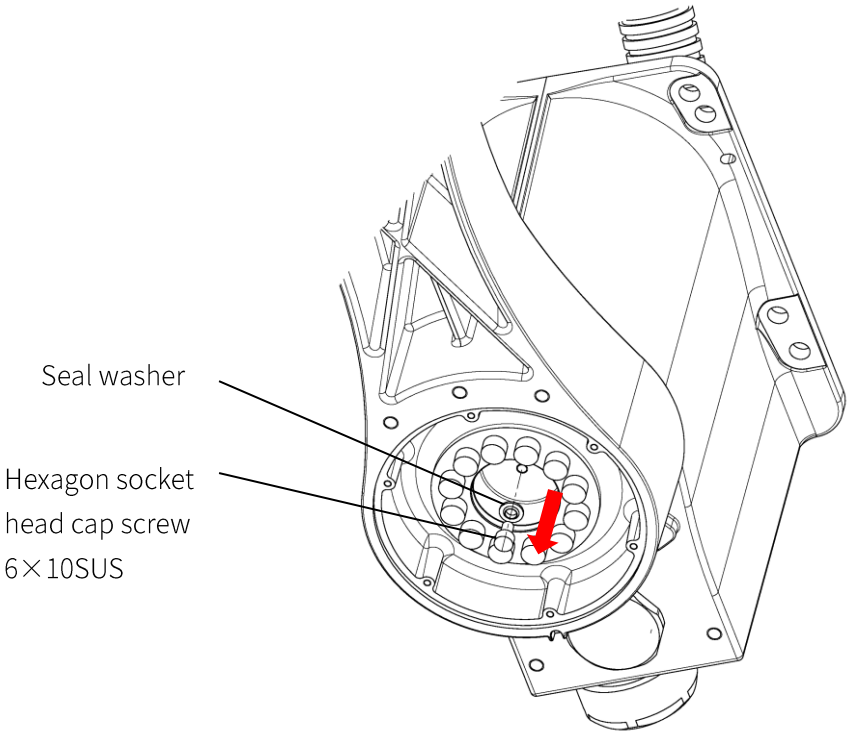



Fig. 6.14 Removing the screw and seal washer

**8 Fit the user-prepared grease nipple (M6).**

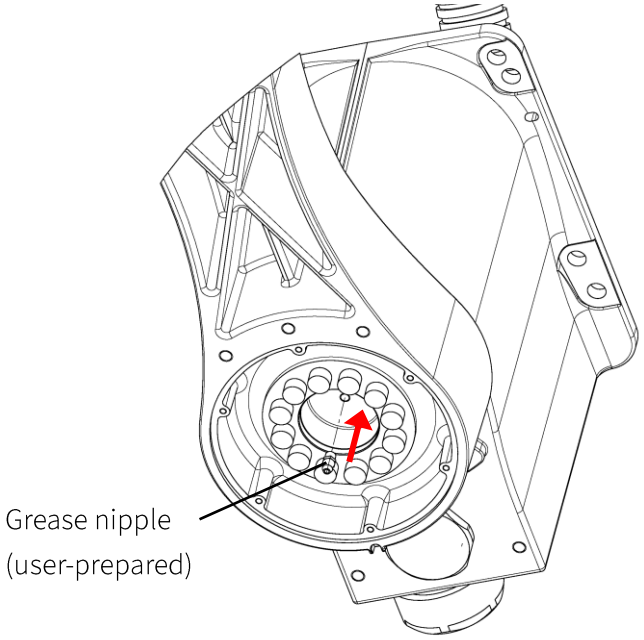



Fig. 6.15 Fitting grease nipple

**9 Remove the hexagon socket head sunk plug (Rc1/8) from the front of the base.**

To release pressure during grease injection, remove the sunk plug from the front of the base.

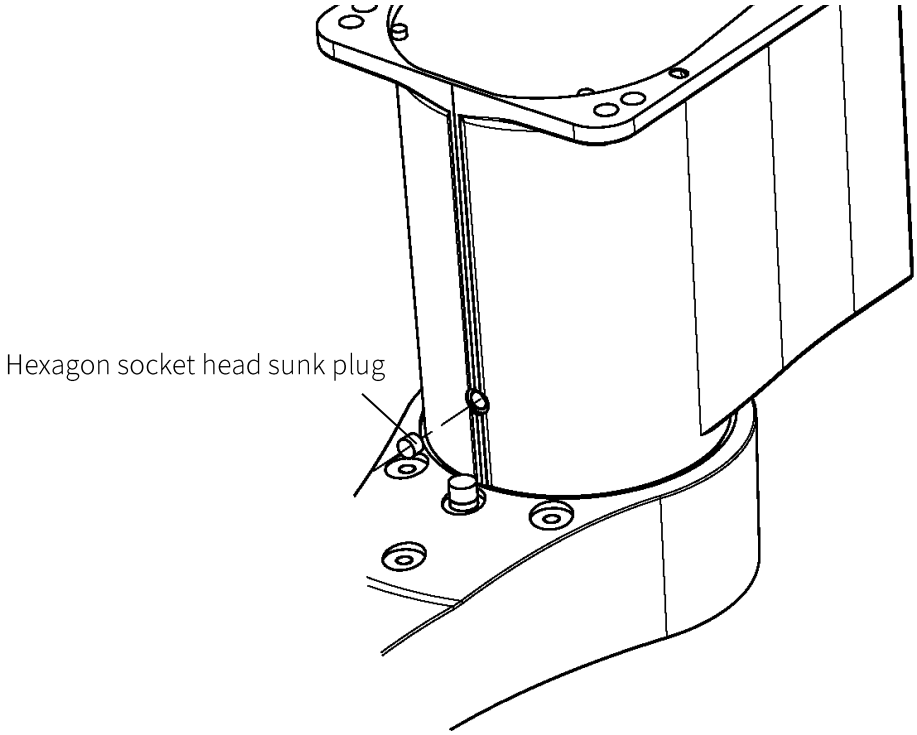



Fig. 6.16 Removing the sunk plug

**10 Fill grease in the arm-1 reduction gear.**

Use the grease gun to fill grease. For the filling amount, see Table 6..

Table 6.1 Filling amount of grease

| Recommended grease | Maker | Filling amount |
|---|---|---|
| Multemp FZ No.00 | Kyodo Yushi Co., Ltd. | 60 g |

**11 Remove the grease nipple.**

When removing, be careful that grease does not leak.

**12 Mount the hexagon socket head cap screw and seal washer in the base.**

Mount the screw and seal washer that were removed in step 7, and be careful that grease does not leak.

**13 Mount the hexagon socket head sunk plug.**

Mount the sunk plug that was removed in step 9. When mounting the plug, wrap seal tape around it so that grease does not leak.

**14 Mount the arm-1 cover to arm 1.**

For the mounting procedure, see “エラー! 参照元が見つかりません。 Arm-1 Cover”.

**15 Move the arm 1 by hand.**

Make sure that it moves smoothly.

**16 Set the zero position of the arm 1.**

For how to set the zero position, see “エラー! 参照元が見つかりません。 エラー! 参照元が見つかりません。” in the “STA1171: THE600 Robot Manual”.

The replacement of the axis-1 reduction gear.

**17 Perform operation at a low speed.**

Perform running-in operation at 20% of overrun for about one hour.

Make sure that there is any problem with operation of each part.

**NOTICE**

**Required**

- Be sure to use the specified types of grease.
  Using grease of other than specified types causes malfunction due to poor lubrication.
- Be sure to observe the filling amount of grease.
  If the filling amount is tool large, the starting torque increases and the internal seal is broken due to rise in the internal pressure in the reduction gear. If the filling amount is too small, malfunction is caused by poor lubrication.

# 7. Robot Zero Point and Position Detector Error

The worker should receive safety training for laborer engaged in industrial robot-related work defined by each country's laws and regulations.
Be aware that SHIBAURA MACHINE's warranty does not cover failures and accidents resulting from the work done by the customer.

## 7.1 Zero Position

This section describes the zero position required for zero setting. When moving the axes 1, 2, and 4 of the robot to the zero position, see "7.1.1 Locations of Zero Point Match-Marks of the Robot" and "STA1171: THE600 Robot Manual", and align the zero point match-mark of each axis. The axis 3 does not have a zero point match-mark. See "エラー! 参照元が見つかりません。 エラー! 参照元が見つかりません。" in the "STA1171: THE600 Robot Manual" to move it to the zero position. The installation of the zero setting tool (optional) is explained in "7.1.2 Installing the Zero Setting Tool (Optional)."

### 7.1.1 Locations of Zero Point Match-Marks of the Robot

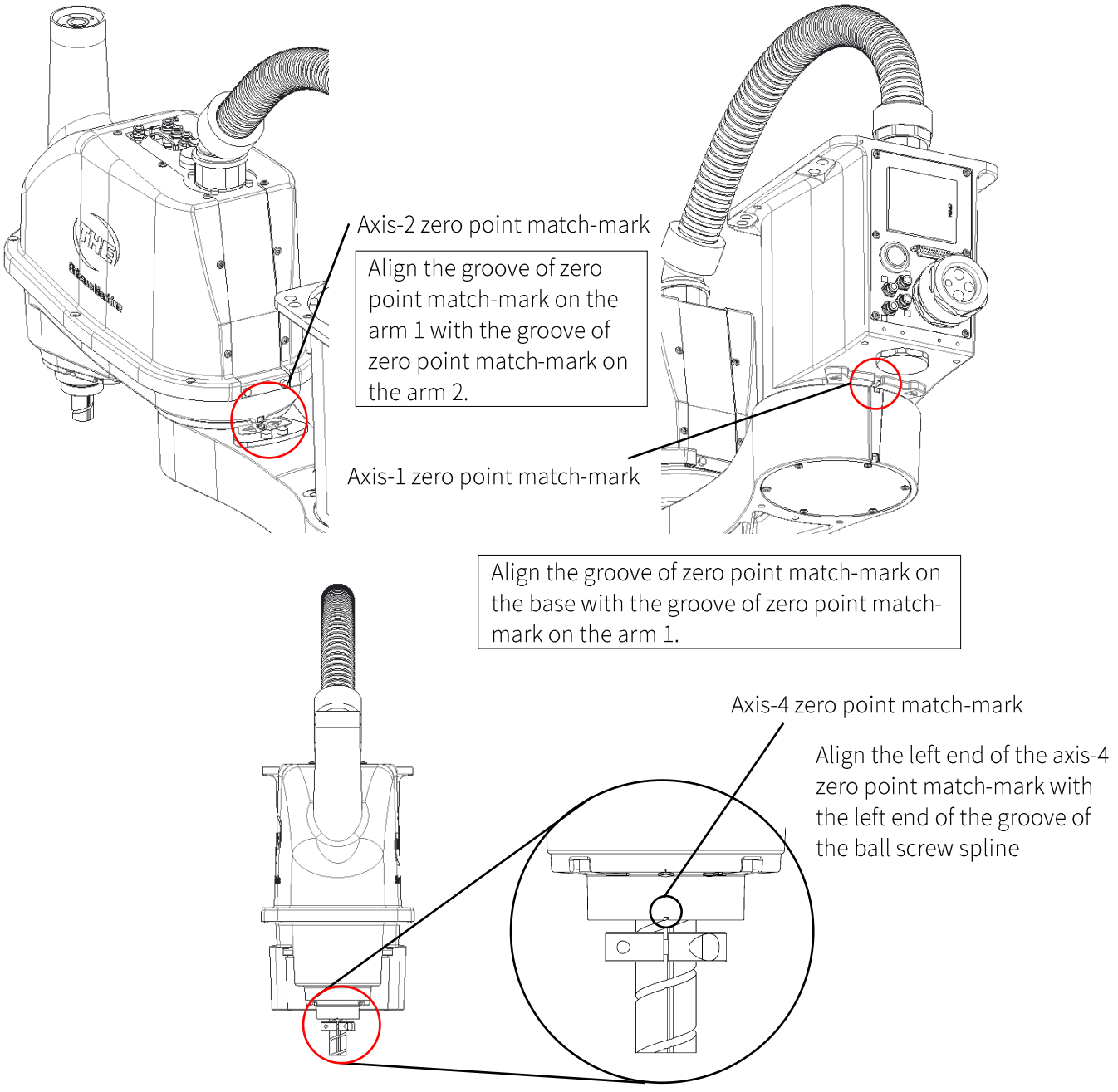



Fig. 7.1 Locations of zero point match-marks

## 7.1.2 Installing the Zero Setting Tool (Optional)

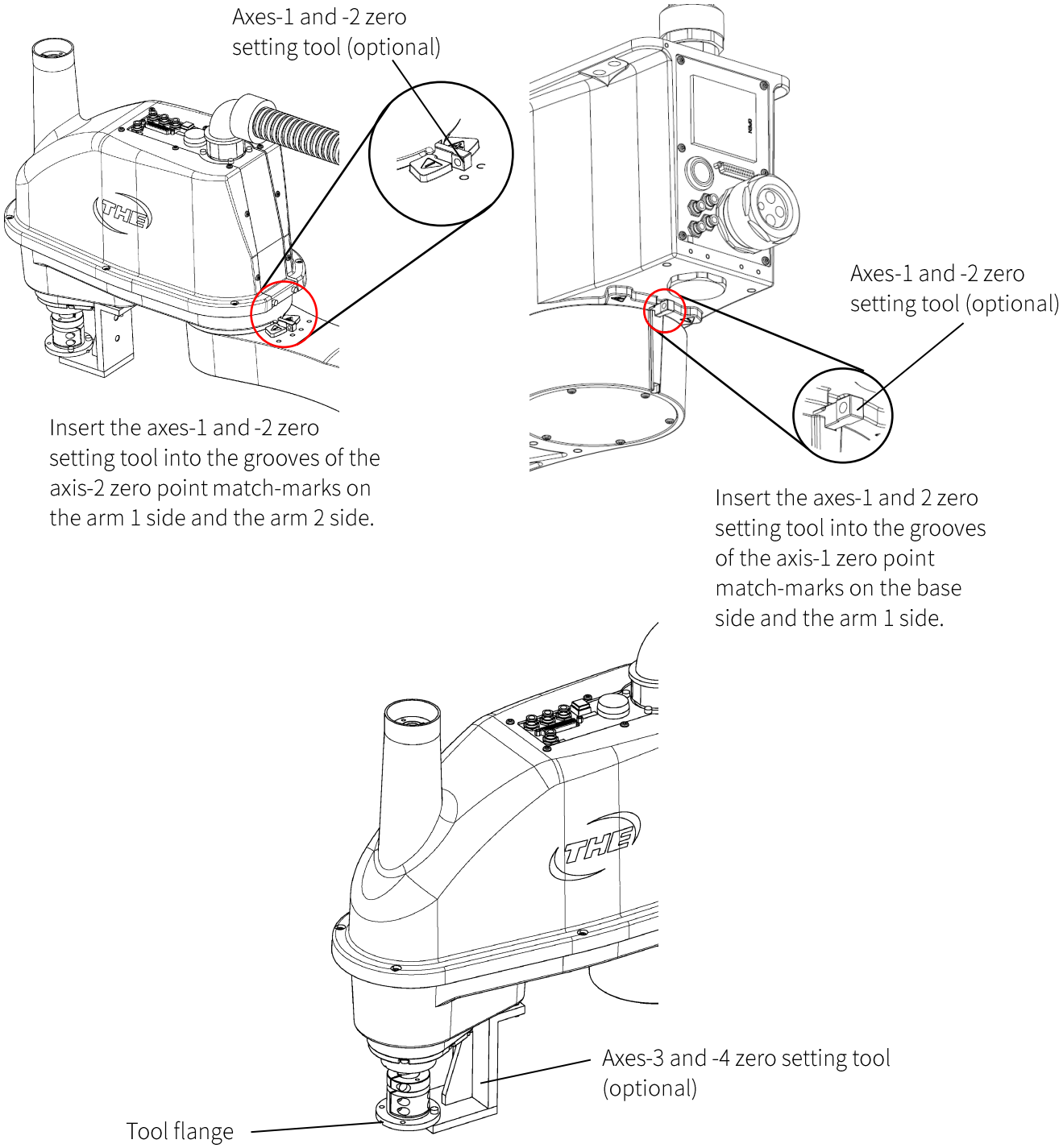



Fig. 7.2 Installing the Zero Setting Tool (Optional)

# 8. Replacement Parts for Maintenance

## 8.1 Notes on Replacement Parts for Maintenance

| DANGER | |
|---|---|
|  **Required** | • Be sure to replace the parts with SHIBAURA MACHINE-specified replacement parts for maintenance.<br>If parts other than the specified parts are used, unpredictable disaster will occur.<br>- Items other than the position detector battery are all special-specification products. When purchasing or ordering them, be sure to contact SHIBAURA MACHINE. |
|  **Caution/ Warning** | • Be sure to contact SHIBAURA MACHINE service company about work other than replacement of the position detector battery. We accepts the work at an extra cost. SHIBAURA MACHINE's warranty does not cover accidents and damage resulting from the replacement performed by the customer and its consequence. |

| NOTICE | |
|---|---|
|  **Required** | • Be sure to replace the parts with SHIBAURA MACHINE-specified replacement parts for maintenance.<br>If parts other than the specified parts are used, the performance and lifespan of the robot will be reduced. The robot will malfunction. |

## 8.2 List of Replacement Parts – Robot

| No | Part name | Model | Drawing No. | Unit code | Manufacturer | Q'ty | Remarks |
|---|---|---|---|---|---|---|---|
| 1 | AC servo motor | | S968416 | Y610D0CW0 | SHIBAURA MACHINE | 1 | Axis 1 |
| 2 | | | S968417 | Y610D0CX0 | | 1 | Axis 2 and Axis 4 |
| 3 | | | S968418 | Y610D0CY0 | | 1 | Axis 3 |
| 5 | Reduction gear | | S968522 | Y610D0CZ0 | SHIBAURA MACHINE | 1 | Axis 1 |
| 6 | | | S981478 | Y610D0D00 | | 1 | Axis 2 |
| 8 | Timing pulley | | S968379 | Y610D0DD0 | SHIBAURA MACHINE | 1 | Axis 3 (ball screw side) |
| 9 | | | S968380 | | | 1 | Axis 3 (motor side) |
| 10 | | | S968381 | | | 1 | Axis 4 (lower side of intermediate shaft) |
| 11 | | | M342404 | | | 1 | Axis 4 (ball screw side) |
| 12 | | | M342405 | | | 1 | Axis 4 (motor side) |
| 13 | | | M342406 | | | 1 | Axis 4 (upper side of intermediate shaft) |
| 14 | Timing belt | | S968393 | | | 1 | Axis 3 |
| 15 | | | S968395 | | | 1 | Axis 4 (ball screw side) |
| 16 | | | S968397 | | | 1 | Axis 4 (motor side) |
| 17 | Ball screw spline unit | | H880457 | Y610D0CP0 | SHIBAURA MACHINE | 1 | |
| 18 | Body harness | | F127810 | Y610D0DF0 | SHIBAURA MACHINE | 1 | |
| 19 | LED Lamp | | M342499 | Y610D0DU0 | SHIBAURA MACHINE | 1 | |
| 20 | O ring | CO0540A | | | NOK Corp. | 1 | Axis 2 |
| 21 | | CO0543A | | | | 1 | Axis 2 |
| 22 | | CO0548A | | | | 1 | Axis 1 |

| No | Part name | Model | Drawing No. | Unit code | Manufacturer | Q'ty | Remarks |
|---|---|---|---|---|---|---|---|
| 23 | Grease | Multemp FZNo.00 | | | Kyodo Yushi Co., Ltd. | | Axis 1 |
| 24 | | SFB No.1 (For reduction gear) | | | Nidec-Shimpo Corp. | | Axis 2 |
| 25 | | AFF grease (For ball screw) | | | THK | | Axis 3 |
| 26 | | LGAF 3E (For key) | | | Japan SKF Co., Ltd. | | All axes |
| 27 | Screw locking adhesive | LOCTITE 241 | | | Henkel Japan Ltd. | | |
| 28 | | LOCTITE 221 | | | | | |
| 29 | Encoder backup battery | Alkaline AA-cell battery | | | | 3 | All axes |
| 30 | Zero setting tool*1 | | M342399 | Y610D0DQ0 | SHIBAURA MACHINE | | Axis 1 and Axis 2 |
| 31 | | | L23752G01 | Y610D0DR0 | | | Axis 3 and Axis 4 |
| 32 | Tension adjustment tool*2 | | L27611G01 | Y610D0DW0 | SHIBAURA MACHINE | | Axis 4 |
| 33 | | | L27612G01 | Y610D0DX0 | | | Axis 3 and Axis 4 |
| 34 | Tool flange | | L12515G01 | Y610D0300 | SHIBAURA MACHINE | | |